IMAT 모의고사 10회분

2023–2026 IMAT의 시험 형식과 출제 범위·유형 경향을 참고해 새로 작성한 모의문제입니다. 실제 기출 문항을 옮겨 적거나 바꿔 쓴 것이 아니라, 같은 형식과 난이도를 목표로 직접 만든 문항입니다.

시험 구성 (2023–2026 공통)

영역	문항 수	문항 번호
Reading skills & general knowledge	4	1–4
Logical reasoning & problem solving	5	5–9
Biology	23	10–32
Chemistry	15	33–47
Physics & Mathematics	13	48–60
합계	**60**	100분

- 채점: 정답 **+1.5**, 오답 **−0.4**, 무응답 **0** (만점 90점)
- 각 문항은 보기 A–E 중 정답 1개 (회차마다 정답이 A–E에 12개씩 고르게 분포 → 찍기로는 점수가 오르지 않음)
- 계산기 사용 불가 → 모든 계산 문항은 손계산이 가능하도록 수치를 맞췄습니다.

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각 파일 끝에 정답표와 한국어 해설이 있습니다. 실전처럼 100분 타이머를 맞추고 풀어 본 뒤 채점하세요.

2023–2026 출제 경향 메모 (작성 시 반영한 부분)

- **Reading/GK**: 짧은 과학·사회 지문의 추론 문제, 그리고 역사·문화·시사 일반상식이 섞여 나옴.
- **Logic**: 삼단논법·가정 찾기, 순서 배열, 비율·속도 문제, 숫자 규칙.
- **Biology**: 세포·분자생물학(복제·전사·번역, 세포 호흡), 유전(멘델·반성유전·가계도), 인체 생리(신경·내분비·순환·신장), 생화학 거대분자 비중이 높음.

- **Chemistry**: 몰 계산·화학양론, 산화수/산화환원, pH·완충, 주기율 경향, 결합·분자구조, 유기 작용기와 이성질체.
- **Physics & Math**: 역학(운동·힘·에너지), 전기회로, 유체·열, 파동, 그리고 대수·로그·삼각함수·확률.

IMAT 모의고사 1회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> For decades, hospital wards relied on fixed visiting hours, partly from a belief that visitors brought infections and disturbed rest. Recent studies of intensive care units that adopted flexible visiting have found no increase in infection rates and, in several cases, a reduction in patient delirium. Critics note that most of these studies were small and that staff workload was rarely measured. Nonetheless, several national health services now recommend that restrictions be justified case by case rather than imposed by default.

1. Which of the following can be most reliably inferred from the passage?
A. The original justification for fixed visiting hours was not supported by the recent studies.
B. Flexible visiting reduces infection rates in intensive care units.
C. National health services have banned fixed visiting hours.
D. Delirium is caused mainly by the absence of visitors.
E. Staff workload increases when visiting hours become flexible.

2. Which statement, if true, would most weaken the critics' position described in the passage?
A. A large multi-centre trial measuring staff workload found no significant change after flexible visiting was introduced.
B. Infection rates in hospitals have fallen over the last decade for unrelated reasons.
C. Delirium is more common in elderly patients.
D. Visiting rules differ between countries.
E. Some patients prefer fewer visitors.

3. Which scientist is credited with first describing the circulation of blood pumped by the heart around the body in the 17th century?
A. Andreas Vesalius
B. Louis Pasteur
C. Edward Jenner
D. Robert Koch
E. William Harvey

4. The Treaty of Rome (1957) is best known for establishing which organisation?
A. The United Nations
B. The World Health Organization
C. NATO
D. The European Economic Community
E. The Council of Europe

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. All surgeons in the hospital are doctors. Some doctors in the hospital speak Italian. No one who speaks Italian works night shifts. Which of the following must be true?

A. Some doctors do not work night shifts.
B. No surgeon works night shifts.
C. All doctors who work night shifts are surgeons.
D. Some surgeons work night shifts.
E. Some surgeons speak Italian.

6. A pharmacy sells a drug at €40 per box. In a promotion, the price is reduced by 25%; after the promotion, the reduced price is raised by 25%. What is the final price?

A. €38.75
B. €37.50
C. €35.00
D. €40.00
E. €42.50

7. Five patients — P, Q, R, S and T — are seen one after another. Q is seen immediately after P. R is seen before P. T is seen last. S is seen after Q. Who is seen second?

A. S
B. Q
C. P
D. T
E. R

8. What is the next number in the sequence: 3, 7, 15, 31, 63, ...?

A. 115
B. 127
C. 128
D. 95
E. 126

9. A tap fills a tank in 6 hours; a drain empties the full tank in 9 hours. If both are open, starting with an empty tank, how long does it take to fill the tank?

A. 3 hours
B. 15 hours
C. 12 hours
D. 7.5 hours
E. 18 hours

Section 3 — Biology (Q10–32)

10. Which organelle is the main site of synthesis of proteins destined for secretion?

A. Smooth endoplasmic reticulum
B. Peroxisome
C. Rough endoplasmic reticulum
D. Free cytosolic ribosomes
E. Lysosome

11. A segment of DNA template strand reads 3'-TACGGA-5'. What is the sequence of the mRNA transcribed from it?

A. 3'-AUGCCU-5'
B. 5'-UACGGA-3'
C. 5'-AUGCCU-3'
D. 5'-ATGCCT-3'
E. 5'-UCCGUA-3'

12. In which phase of mitosis do sister chromatids separate and move towards opposite poles?

A. Anaphase
B. Prometaphase
C. Metaphase
D. Prophase
E. Telophase

13. Which of the following processes produces the greatest number of ATP molecules per glucose in aerobic respiration?

A. Glycolysis
B. Krebs cycle (substrate-level phosphorylation)
C. Lactic acid fermentation
D. Link reaction
E. Oxidative phosphorylation

14. Red–green colour blindness is an X-linked recessive trait. A woman with normal vision whose father was colour-blind has children with a man with normal vision. What is the probability that their first son is colour-blind?

A. 3/4
B. 0
C. 1/4
D. 1/2
E. 1

15. Which molecule acts as the final electron acceptor in the mitochondrial electron transport chain?

A. Water
B. Cytochrome c
C. NAD^+
D. Oxygen
E. FAD

16. Which of the following is NOT a function of the liver?

A. Secretion of insulin
B. Synthesis of albumin
C. Production of bile
D. Storage of glycogen
E. Deamination of amino acids

17. Which statement about enzymes is correct?

A. Non-competitive inhibitors always bind to the active site.

B. A competitive inhibitor binds to the active site and its effect can be overcome by increasing substrate concentration.

C. Enzymes increase the activation energy of a reaction.

D. Enzymes change the equilibrium constant of a reaction.

E. Enzymes are consumed during the reaction.

18. During the depolarisation phase of a neuronal action potential, which event occurs?

A. The Na^+/K^+ pump moves 3 Na^+ in and 2 K^+ out.

B. The membrane potential becomes more negative.

C. K^+ rushes into the cell through voltage-gated channels.

D. Cl^- leaves the cell.

E. Na^+ rushes into the cell through voltage-gated channels.

19. Which blood vessel carries oxygenated blood from the lungs to the heart?

A. Coronary sinus

B. Pulmonary vein

C. Aorta

D. Pulmonary artery

E. Superior vena cava

20. In a dihybrid cross AaBb × AaBb with independent assortment and complete dominance, what fraction of offspring show both dominant phenotypes?

A. 3/16

B. 12/16

C. 9/16

D. 1/16

E. 6/16

21. Which hormone is released by the posterior pituitary?

A. Prolactin

B. Thyroid-stimulating hormone

C. Follicle-stimulating hormone

D. Antidiuretic hormone (ADH)

E. Growth hormone

22. Which type of bond holds the two strands of the DNA double helix together?

A. Hydrogen bonds between bases

B. Disulfide bonds

C. Glycosidic bonds

D. Phosphodiester bonds

E. Peptide bonds

23. Which of the following cells is responsible for producing antibodies?

A. Cytotoxic T cells

B. Plasma cells

C. Macrophages

D. Neutrophils

E. Helper T cells

24. In the nephron, where does most reabsorption of glucose normally occur?

A. Proximal convoluted tubule
B. Descending limb of the loop of Henle
C. Distal convoluted tubule
D. Collecting duct
E. Glomerulus

25. Which statement about meiosis is correct?

A. Crossing over occurs during prophase II.
B. It produces two genetically identical diploid cells.
C. It occurs in all somatic cells.
D. Homologous chromosomes separate during anaphase I.
E. DNA replicates between meiosis I and meiosis II.

26. Which of these is a polysaccharide used for energy storage in animals?

A. Sucrose
B. Chitin
C. Starch
D. Glycogen
E. Cellulose

27. A mutation changes a codon from UAC (Tyr) to UAA. What type of mutation is this?

A. Frameshift mutation
B. Insertion
C. Silent mutation
D. Missense mutation
E. Nonsense mutation

28. Which component of the plasma membrane is mainly responsible for regulating its fluidity in animal cells?

A. Integral proteins
B. Cellulose
C. Cholesterol
D. Glycolipids
E. Glycoproteins

29. Bacteria differ from eukaryotic cells because bacteria:

A. cannot contain plasmids.
B. always lack a cell wall.
C. have 70S ribosomes and lack membrane-bound organelles.
D. lack ribosomes.
E. have a nucleus surrounded by a double membrane.

30. Where in the human digestive system does most absorption of nutrients take place?

A. Large intestine
B. Stomach
C. Oesophagus
D. Duodenum only
E. Small intestine (jejunum and ileum)

31. Which of the following describes the role of tRNA in translation?

A. It brings a specific amino acid to the ribosome and pairs its anticodon with an mRNA codon.
B. It carries the genetic code from the nucleus to the cytoplasm.
C. It forms the structural core of the ribosome.
D. It removes introns from pre-mRNA.
E. It catalyses formation of phosphodiester bonds.

32. In a population in Hardy–Weinberg equilibrium, the frequency of a recessive phenotype is 0.16. What is the frequency of heterozygotes?

A. 0.24
B. 0.48
C. 0.60
D. 0.36
E. 0.16

Section 4 — Chemistry (Q33–47)

33. How many moles of oxygen atoms are present in 0.5 mol of $Ca(NO_3)_2$?

A. 2
B. 3
C. 1.5
D. 6
E. 9

34. What is the oxidation number of manganese in $KMnO_4$?

A. +5
B. +4
C. +2
D. +7
E. +6

35. What is the pH of a 0.001 mol/L solution of HCl at 25 °C?

A. 2
B. 1
C. 13
D. 11
E. 3

36. Which element has the highest first ionisation energy?

A. Ar
B. Cl
C. K
D. Mg
E. Na

37. What is the molecular geometry of NH_3?

A. Bent
B. Trigonal planar
C. Trigonal pyramidal
D. Linear
E. Tetrahedral

38. Balance the equation: $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$. What is the coefficient of O_2 when the smallest whole-number coefficients are used?

A. 7
B. 5
C. 3
D. 10
E. 4

39. Which of the following compounds contains an ester functional group?

A. CH_3CH_2OH
B. CH_3COOCH_3
C. CH_3CHO
D. CH_3COCH_3
E. CH_3COOH

40. 20 g of NaOH (M = 40 g/mol) is dissolved in water to make 500 mL of solution. What is the concentration?

A. 4 mol/L
B. 1 mol/L
C. 0.5 mol/L
D. 2 mol/L
E. 0.25 mol/L

41. Which type of intermolecular force is mainly responsible for the high boiling point of water compared with H_2S?

A. Ionic bonding
B. Covalent bonding
C. London dispersion forces
D. Hydrogen bonding
E. Metallic bonding

42. Which of the following is a buffer solution?

A. $HNO_3 + KNO_3$
B. HCl + NaCl
C. $CH_3COOH + CH_3COONa$
D. NaOH + NaCl
E. $H_2SO_4 + Na_2SO_4$

43. For the exothermic equilibrium $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, which change increases the yield of NH_3?

A. Adding a catalyst
B. Increasing pressure
C. Increasing temperature
D. Removing N_2
E. Decreasing pressure

44. How many structural isomers have the formula C_4H_{10}?

A. 1
B. 5
C. 3
D. 2
E. 4

45. An isotope has 17 protons and 20 neutrons. Which is it?

A. ^{37}Cl
B. ^{37}Ar
C. ^{34}S
D. ^{20}Ca
E. ^{37}K

46. In the reaction $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, which species is the oxidising agent?

A. There is no oxidising agent.
B. Cu
C. Cu^{2+}
D. Zn
E. Zn^{2+}

47. Which of the following alkenes can show cis–trans (geometric) isomerism?

A. Ethene
B. Propene
C. 2-methylpropene
D. But-1-ene
E. But-2-ene

Section 5 — Physics & Mathematics (Q48–60)

48. A car accelerates uniformly from rest to 20 m/s in 5 s. How far does it travel in this time?

A. 200 m
B. 40 m
C. 20 m
D. 50 m
E. 100 m

49. A 2 kg object is lifted vertically by 3 m. Taking g = 10 m/s², what is the gain in gravitational potential energy?

A. 6 J
B. 30 J
C. 600 J
D. 60 J
E. 15 J

50. Two resistors of 6 Ω and 3 Ω are connected in parallel. What is the equivalent resistance?

A. 9 Ω
B. 0.5 Ω
C. 18 Ω
D. 4.5 Ω
E. 2 Ω

51. A wave has frequency 50 Hz and wavelength 4 m. What is its speed?

A. 200 m/s
B. 100 m/s
C. 12.5 m/s
D. 46 m/s
E. 54 m/s

52. What is the pressure exerted by a column of water 10 m deep (ρ = 1000 kg/m³, g = 10 m/s²), excluding atmospheric pressure?

A. 10^4 Pa
B. 10^3 Pa
C. 10^7 Pa
D. 10^6 Pa
E. 10^5 Pa

53. A body of mass 4 kg moving at 3 m/s has kinetic energy of:

A. 24 J
B. 36 J
C. 6 J
D. 12 J
E. 18 J

54. Which of the following is the SI unit of power?

A. kg·m²/s²
B. kg·m/s²
C. kg·m²/s³
D. kg·m/s
E. N·s

55. Solve for x: $\log_2(x) + \log_2(x - 2) = 3$.

A. 4
B. −2 and 4
C. 8
D. −2
E. 2

56. What is the value of $\sin(30°) \cdot \cos(60°) + \cos(30°) \cdot \sin(60°)$?

A. $\sqrt{3}/2$
B. 1
C. 1/2
D. 0
E. $\sqrt{3}$

57. Two fair six-sided dice are rolled. What is the probability that the sum is 7?

A. 1/9
B. 1/12
C. 7/36
D. 1/6
E. 1/36

58. What is the area of a circle whose circumference is 10π cm?

A. 5π cm²
B. 50π cm²
C. 25π cm²
D. 100π cm²
E. 10π cm²

59. The solutions of $x^2 - 5x + 6 = 0$ are:

A. $x = -2$ and $x = -3$
B. $x = 1$ and $x = 6$
C. $x = 2$ and $x = 3$
D. $x = -1$ and $x = 6$
E. $x = 5$ and $x = 6$

60. A current of 2 A flows through a resistor for 10 s. How much charge passes through it?

A. 5 C
B. 0.2 C
C. 12 C
D. 200 C
E. 20 C

정답 및 해설

번호	정답	해설
1	A	감염 증가가 없었다는 결과는 "방문객이 감염을 옮긴다"는 기존 근거를 지지하지 않음. B는 "감소"까지 주장해 과도.
2	A	비판은 "연구 규모가 작고 업무량 미측정". 대규모·업무량 측정 연구가 차이 없음을 보이면 비판이 약화.
3	E	윌리엄 하비(1628, *De Motu Cordis*).
4	D	로마조약 → 유럽경제공동체(EEC) 설립.
5	A	이탈리아어를 하는 의사는 야간근무 안 함 → 야간근무 안 하는 의사가 존재.
6	B	40×0.75=30, 30×1.25=37.5.
7	C	T 마지막, R<P, P 바로 뒤 Q, S는 Q 뒤 → R, P, Q, S, T. 두 번째는 P.
8	B	각 항 = 2×이전+1 → 127.
9	E	1/6 − 1/9 = 1/18 → 18시간.
10	C	분비 단백질은 조면소포체 리보솜에서 합성.
11	C	주형 3'-TACGGA-5' → mRNA 5'-AUGCCU-3'.
12	A	후기(anaphase)에 자매염색분체 분리.
13	E	산화적 인산화가 ATP 대부분(~26–28개) 생성.
14	D	어머니는 보인자(X^AX^a) → 아들 1/2.
15	D	최종 전자수용체는 O_2.
16	A	인슐린은 췌장 β세포.
17	B	경쟁적 저해는 기질 농도↑로 극복 가능.
18	E	탈분극: 전압개폐 Na^+ 통로로 Na^+ 유입.
19	B	폐정맥이 산소 풍부한 혈액을 좌심방으로.
20	C	9:3:3:1 → 9/16.
21	D	후엽: ADH, 옥시토신.
22	A	두 가닥은 염기 간 수소결합.
23	B	형질세포(분화된 B세포)가 항체 분비.
24	A	포도당은 근위세뇨관에서 거의 전부 재흡수.
25	D	감수 1분열 후기에 상동염색체 분리.
26	D	동물의 저장 다당류 = 글리코겐.
27	E	UAA는 종결코돈 → 넌센스 돌연변이.
28	C	콜레스테롤이 막 유동성 조절.
29	C	원핵세포: 70S 리보솜, 막성 세포소기관 없음.
30	E	소장(공장·회장)에서 대부분 흡수.
31	A	tRNA는 아미노산 운반, 안티코돈-코돈 결합.
32	B	$q^2=0.16$ → q=0.4, p=0.6, 2pq=0.48.
33	B	1 mol당 O 6개 → 0.5×6=3.
34	D	K(+1)+Mn+4(−2)=0 → +7.

번호	정답	해설
35	E	$[H^+]=10^{-3}$ → pH 3.
36	A	비활성기체 Ar이 가장 높음.
37	C	비공유전자쌍 1개 → 삼각뿔.
38	B	$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$.
39	B	아세트산메틸(에스터).
40	B	0.5 mol / 0.5 L = 1 mol/L.
41	D	물의 수소결합.
42	C	약산 + 그 짝염기.
43	B	기체 몰수 4→2, 압력↑ 시 정반응. 촉매는 수율 불변.
44	D	부탄, 2-메틸프로페인.
45	A	양성자 17 = Cl, 질량수 37.
46	C	Cu^{2+}가 환원됨 → 산화제.
47	E	부트-2-엔만 각 이중결합 탄소에 서로 다른 치환기.
48	D	평균속도 10 m/s × 5 s = 50 m.
49	D	$mgh = 2\times10\times3 = 60$ J.
50	E	$6\cdot3/(6+3) = 2\ \Omega$.
51	A	$v = f\lambda = 200$ m/s.
52	E	$\rho gh = 1000\times10\times10 = 10^5$ Pa.
53	E	$\frac{1}{2}\cdot4\cdot9 = 18$ J.
54	C	$W = J/s = kg\cdot m^2/s^3$.
55	A	$x(x-2)=8 \rightarrow x=4$ ($x=-2$는 정의역 밖).
56	B	$\sin(30°+60°) = \sin90° = 1$.
57	D	6/36 = 1/6.
58	C	$r = 5 \rightarrow 25\pi$.
59	C	$(x-2)(x-3)=0$.
60	E	$Q = It = 20$ C.

IMAT 모의고사 2회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Antibiotic resistance is often described as a problem of overuse in human medicine. Yet in many countries more antibiotics are given to farm animals than to people, frequently to promote growth rather than to treat disease. Resistant bacteria and resistance genes can move from animals to humans through food, water and direct contact. Some countries that restricted growth-promoting antibiotics saw resistance in animal bacteria fall within a few years, although effects on human infections were harder to measure.

1. What is the main point of the passage?
A. Resistance genes cannot transfer between species.
B. Restricting antibiotics in farming has eliminated resistance in humans.
C. Agricultural use of antibiotics is a significant, and sometimes overlooked, contributor to resistance.
D. Human overuse of antibiotics is not a problem.
E. Antibiotic resistance is caused only by agriculture.

2. According to the passage, what happened after some countries restricted growth-promoting antibiotics?
A. Antibiotic use in humans increased.
B. Resistance genes stopped moving through water.
C. Human infections immediately decreased.
D. Farm productivity collapsed.
E. Resistance in bacteria from animals declined.

3. Which of the following Italian cities was the capital of the Kingdom of Italy immediately before Rome became capital in 1871?
A. Venice
B. Naples
C. Turin
D. Milan
E. Florence

4. The discovery of the structure of DNA in 1953 is associated with Watson and Crick, but which scientist's X-ray diffraction images were crucial to it?
A. Marie Curie
B. Rosalind Franklin
C. Lise Meitner
D. Dorothy Hodgkin
E. Barbara McClintock

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "If a patient has a fever, the doctor orders a blood test. The doctor did not order a blood test." Which conclusion follows?

A. The doctor made a mistake.
B. The patient has an infection.
C. The patient has a fever.
D. Nothing can be concluded.
E. The patient does not have a fever.

6. A train leaves at 08:45 and travels 270 km at an average speed of 120 km/h. At what time does it arrive?

A. 10:55
B. 11:30
C. 11:15
D. 10:45
E. 11:00

7. In a class of 30 students, 18 study biology, 14 study chemistry and 5 study neither. How many study both?

A. 9
B. 7
C. 11
D. 3
E. 5

8. Which number completes the pattern: 2, 6, 12, 20, 30, __?

A. 48
B. 36
C. 42
D. 40
E. 44

9. A medicine must be taken every 8 hours. The first dose is taken at 22:00 on Monday. When is the 5th dose taken?

A. 22:00 Tuesday
B. 06:00 Tuesday
C. 22:00 Wednesday
D. 06:00 Wednesday
E. 14:00 Wednesday

Section 3 — Biology (Q10–32)

10. Which of the following is found in plant cells but not in animal cells?

A. Chloroplasts
B. Ribosomes
C. Golgi apparatus
D. Mitochondria
E. Plasma membrane

11. The enzyme that joins Okazaki fragments during DNA replication is:
A. DNA ligase
B. DNA polymerase III
C. Primase
D. Topoisomerase
E. Helicase

12. Which statement about the cell cycle is correct?
A. Mitosis takes up most of the cell cycle.
B. DNA is replicated during G1.
C. Cytokinesis occurs during prophase.
D. Cells in G0 are actively dividing.
E. DNA is replicated during S phase.

13. How many net ATP molecules are produced directly by glycolysis per molecule of glucose?
A. 2
B. 1
C. 32
D. 4
E. 38

14. A man with blood group AB and a woman with blood group O have a child. Which statement is correct?
A. The child may have blood group O.
B. The child must have blood group AB.
C. The child may have blood group AB.
D. The child can only have blood group A or B.
E. The child must have blood group A.

15. Where does the Calvin cycle take place?
A. Stroma of the chloroplast
B. Mitochondrial matrix
C. Thylakoid membrane
D. Nucleus
E. Cytoplasm

16. Which structure prevents food from entering the trachea during swallowing?
A. Pharynx
B. Soft palate
C. Larynx
D. Uvula
E. Epiglottis

17. Which of these molecules is a steroid hormone?
A. Oxytocin
B. Cortisol
C. Glucagon
D. Insulin
E. Adrenaline

18. Which ion entering the presynaptic terminal triggers neurotransmitter release?

A. Na^+
B. K^+
C. Mg^{2+}
D. Cl^-
E. Ca^{2+}

19. In which chamber of the heart does blood have the highest pressure during systole?

A. Right ventricle
B. Left ventricle
C. Left atrium
D. All chambers have equal pressure
E. Right atrium

20. A test cross of a plant with dominant phenotype gives 50% dominant and 50% recessive offspring. What is the genotype of the tested plant?

A. aa
B. Cannot be determined
C. AA or Aa equally likely
D. Aa
E. AA

21. Which hormone stimulates the conversion of glycogen to glucose in the liver?

A. Calcitonin
B. Aldosterone
C. Glucagon
D. Melatonin
E. Insulin

22. Which base is found in RNA but not in DNA?

A. Guanine
B. Adenine
C. Uracil
D. Cytosine
E. Thymine

23. Vaccination mainly works by:

A. providing passive immunity through antibodies from another organism.
B. preventing pathogens from entering the body.
C. stimulating the production of memory cells.
D. directly killing pathogens.
E. increasing the number of red blood cells.

24. The hormone aldosterone primarily acts on the kidney to:

A. decrease Na^+ reabsorption.
B. increase water excretion.
C. increase Na^+ reabsorption and K^+ secretion.
D. increase urea synthesis.
E. increase glucose reabsorption.

25. A human cell in metaphase of mitosis contains how many chromatids?
A. 69
B. 46
C. 92
D. 184
E. 23

26. Which level of protein structure is defined by the sequence of amino acids?
A. Quaternary
B. Domain
C. Secondary
D. Primary
E. Tertiary

27. The deletion of a single nucleotide in a coding region usually causes:
A. a change in the promoter.
B. removal of one amino acid only.
C. no change in the protein.
D. a frameshift that alters all downstream codons.
E. a single amino acid substitution.

28. Which transport process requires ATP directly?
A. Primary active transport
B. Simple diffusion
C. Diffusion through channels
D. Facilitated diffusion
E. Osmosis

29. Viruses are not considered cells because they:
A. have proteins.
B. contain nucleic acids.
C. can mutate.
D. cannot reproduce outside a host cell and lack their own metabolism.
E. infect bacteria.

30. Bile helps digestion by:
A. absorbing vitamin B12.
B. digesting starch.
C. emulsifying fats.
D. hydrolysing proteins.
E. neutralising stomach acid only.

31. Which codon serves as the start codon in most organisms?
A. AUG
B. UAA
C. GUG
D. UAG
E. UGA

32. Which of the following is an example of natural selection?

A. Giraffes stretching their necks to reach leaves and passing longer necks to offspring.
B. A person becoming stronger by training.
C. Producing hybrid corn.
D. Bacteria surviving antibiotic treatment because of a resistance gene, then multiplying.
E. Breeding dogs for herding.

Section 4 — Chemistry (Q33–47)

33. What is the molar mass of H_2SO_4? (H = 1, S = 32, O = 16)

A. 49 g/mol
B. 96 g/mol
C. 100 g/mol
D. 64 g/mol
E. 98 g/mol

34. What is the oxidation number of chromium in $Cr_2O_7^{2-}$?

A. +7
B. +6
C. +12
D. +3
E. +4

35. What is the pH of a 0.01 mol/L NaOH solution at 25 °C?

A. 10
B. 2
C. 7
D. 14
E. 12

36. Down group 1, from Li to Cs, which property increases?

A. Number of valence electrons
B. Melting point
C. Electronegativity
D. First ionisation energy
E. Atomic radius

37. Which molecule is non-polar despite having polar bonds?

A. SO_2
B. NH_3
C. CO_2
D. H_2O
E. HCl

38. How many grams of water are produced when 4 g of H_2 reacts completely with excess O_2? (H = 1, O = 16)

A. 18 g
B. 9 g
C. 36 g
D. 32 g
E. 72 g

39. Oxidation of a primary alcohol under strong conditions gives:

A. a tertiary alcohol.
B. a carboxylic acid.
C. an ether.
D. an alkene.
E. a ketone.

40. What volume of 2 mol/L HCl is needed to prepare 500 mL of 0.4 mol/L HCl?

A. 50 mL
B. 100 mL
C. 200 mL
D. 400 mL
E. 250 mL

41. Which bond type forms when electrons are transferred from one atom to another?

A. Polar covalent
B. Hydrogen
C. Non-polar covalent
D. Ionic
E. Metallic

42. What is the conjugate base of $H_2PO_4^-$?

A. H_3PO_4
B. PO_4^{3-}
C. OH^-
D. HPO_4^{2-}
E. H_2O

43. Increasing temperature increases the rate of a reaction mainly because:

A. more molecules have energy greater than the activation energy.
B. the activation energy decreases.
C. the reaction becomes exothermic.
D. the concentration of reactants increases.
E. the equilibrium constant always increases.

44. Which compound is an aldehyde?

A. $CH_3CH_2NH_2$
B. CH_3CH_2CHO
C. $CH_3CH_2OCH_3$
D. CH_3COCH_3
E. CH_3CH_2COOH

45. What is the electron configuration of a neutral oxygen atom?
A. $1s^2\ 2s^2\ 2p^4$
B. $1s^2\ 2s^2\ 2p^2$
C. $1s^2\ 2s^2\ 2p^6$
D. $1s^2\ 2p^6$
E. $1s^2\ 2s^4\ 2p^2$

46. In an electrolytic cell, oxidation occurs at the:
A. both electrodes equally.
B. anode.
C. cathode.
D. salt bridge.
E. electrolyte.

47. Which of the following is an amino acid functional group pair?
A. Thiol and nitrile
B. Amine and carboxyl
C. Ketone and ester
D. Ether and amide
E. Hydroxyl and aldehyde

Section 5 — Physics & Mathematics (Q48–60)

48. A ball is dropped from rest. Ignoring air resistance ($g = 10\ m/s^2$), what is its speed after 3 s?
A. 15 m/s
B. 3 m/s
C. 45 m/s
D. 30 m/s
E. 10 m/s

49. A force of 20 N moves an object 5 m in the direction of the force. How much work is done?
A. 4 J
B. 100 J
C. 15 J
D. 200 J
E. 25 J

50. Three 6 Ω resistors are connected in series to a 9 V battery. What current flows?
A. 1.5 A
B. 0.5 A
C. 4.5 A
D. 3 A
E. 1 A

51. The speed of light in vacuum is approximately 3×10^8 m/s. What is the wavelength of light with frequency 6×10^{14} Hz?

A. 5×10^{-7} m
B. 2×10^{-6} m
C. 5×10^{-6} m
D. 1.8×10^{23} m
E. 2×10^{6} m

52. An ideal gas at constant temperature has its volume halved. Its pressure:

A. quadruples.
B. stays the same.
C. becomes zero.
D. halves.
E. doubles.

53. A 1000 kg car moves at 10 m/s. What is its momentum?

A. 50 000 kg·m/s
B. 100 000 kg·m/s
C. 1000 kg·m/s
D. 100 kg·m/s
E. 10 000 kg·m/s

54. What is the power of a device that uses 1200 J in 1 minute?

A. 20 W
B. 72 000 W
C. 1200 W
D. 10 W
E. 60 W

55. If $3^x = 81$, what is x?

A. 4
B. 9
C. 81
D. 3
E. 27

56. In a right triangle, the legs are 6 cm and 8 cm. What is the hypotenuse?

A. 12 cm
B. 48 cm
C. 10 cm
D. 7 cm
E. 14 cm

57. A bag contains 3 red and 5 blue balls. Two are drawn without replacement. What is the probability that both are red?

A. 3/8
B. 9/64
C. 3/28
D. 3/14
E. 1/4

58. What is the slope of the line through (1, 2) and (4, 11)?

A. 2
B. 13/5
C. 9
D. 3
E. 1/3

59. Simplify $(x^2 - 9)/(x - 3)$ for $x \neq 3$.

A. $x + 9$
B. $3x$
C. $x - 3$
D. $x + 3$
E. $x^2 - 3$

60. An object of mass 5 kg experiences a net force of 15 N. What is its acceleration?

A. 3 m/s²
B. 75 m/s²
C. 10 m/s²
D. 0.33 m/s²
E. 20 m/s²

정답 및 해설

번호	정답	해설
1	C	농업 사용도 내성의 주요(간과된) 원인이라는 것이 요지.
2	E	동물 세균의 내성이 몇 년 안에 감소. 인체 효과는 측정 어려움.
3	E	피렌체(1865–1871).
4	B	로절린드 프랭클린의 X선 회절 사진(Photo 51).
5	E	대우(modus tollens): 검사 안 함 → 열 없음.
6	E	270/120 = 2.25 h = 2시간 15분 → 08:45 + 2:15 = 11:00.
7	B	30−5=25 = 18+14−x → x=7.
8	C	n(n+1): 6×7=42.
9	D	22:00 월 → 06:00 화 → 14:00 화 → 22:00 화 → 06:00 수(5번째).
10	A	엽록체(그 외 세포벽, 큰 액포).
11	A	DNA 리가아제.
12	E	S기에 DNA 복제.
13	A	4개 생성 − 2개 소모 = 순 2 ATP.
14	D	AB × OO → A 또는 B만 가능, O와 AB 불가.
15	A	엽록체 스트로마.
16	E	후두개(epiglottis).
17	B	코르티솔(부신피질, 스테로이드).
18	E	Ca^{2+} 유입 → 시냅스 소포 방출.
19	B	좌심실(체순환 압력).
20	D	1:1 → 이형접합 Aa.
21	C	글루카곤 → 글리코겐 분해.
22	C	우라실.
23	C	기억세포 형성(능동면역).
24	C	알도스테론: Na^+ 재흡수↑, K^+ 분비↑.
25	C	46 염색체 × 2 염색분체 = 92.
26	D	1차 구조 = 아미노산 서열.
27	D	틀이동 돌연변이.
28	A	1차 능동수송(예: Na^+/K^+-ATPase).
29	D	숙주 밖 증식 불가, 독자적 대사 없음.
30	C	지방 유화.
31	A	AUG.
32	D	변이 존재 → 선택압 → 생존·번식. A는 라마르크설.
33	E	2+32+64 = 98.
34	B	2x + 7(−2) = −2 → x = +6.
35	E	pOH 2 → pH 12.

번호	정답	해설
36	E	원자 반지름 증가.
37	C	CO_2 직선형 → 쌍극자 상쇄.
38	C	4 g H_2 = 2 mol → H_2O 2 mol = 36 g.
39	B	1차 알코올 → 알데하이드 → 카복실산.
40	B	$C_1V_1=C_2V_2$ → V_1 = 0.4×500/2 = 100 mL.
41	D	이온결합.
42	D	H^+ 하나 제거 → HPO_4^{2-}.
43	A	활성화에너지 이상 분자 비율 증가.
44	B	프로판알(–CHO).
45	A	8개 전자: $1s^2\ 2s^2\ 2p^4$.
46	B	산화는 항상 anode.
47	B	아미노기 + 카복실기.
48	D	v = gt = 30 m/s.
49	B	W = Fd = 100 J.
50	B	18 Ω → 9/18 = 0.5 A.
51	A	$\lambda = c/f = 3\times10^8/6\times10^{14} = 5\times10^{-7}$ m.
52	E	보일 법칙.
53	E	p = mv = 10 000.
54	A	1200/60 = 20 W.
55	A	$3^4 = 81$.
56	C	6-8-10.
57	C	3/8 × 2/7 = 6/56 = 3/28.
58	D	(11−2)/(4−1) = 3.
59	D	(x−3)(x+3)/(x−3).
60	A	$a = F/m = 3\ m/s^2$.

IMAT 모의고사 3회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Sleep researchers once treated dreaming as a by-product of random brain activity. More recent work suggests that sleep, particularly slow-wave sleep, plays an active role in consolidating memories: brain regions active during learning are reactivated during subsequent sleep. In experiments, volunteers who slept after learning a task outperformed those who stayed awake for the same period. However, whether dreams themselves contribute to this process, or merely accompany it, remains disputed.

1. Which of the following is presented in the passage as still uncertain?

A. Whether volunteers who slept performed better.
B. Whether slow-wave sleep exists.
C. Whether brain regions are reactivated during sleep.
D. Whether sleep helps memory consolidation.
E. Whether dreams play a causal role in memory consolidation.

2. The experiment described compares volunteers who slept with volunteers who stayed awake "for the same period". Why is this detail important?

A. It shows that waking impairs the brain.
B. It controls for the time elapsed between learning and testing.
C. It proves dreams are random.
D. It ensures that both groups dreamed.
E. It ensures that both groups learned different tasks.

3. Which disease was declared eradicated worldwide by the World Health Organization in 1980?

A. Measles
B. Smallpox
C. Malaria
D. Polio
E. Tuberculosis

4. Who wrote *The Divine Comedy*?

A. Ariosto
B. Machiavelli
C. Dante Alighieri
D. Petrarch
E. Boccaccio

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "Only students who pass the exam can enrol in the course. Marta enrolled in the course." What can be concluded?

A. All students who passed enrolled.
B. Marta did not take the exam.
C. Nothing can be concluded.
D. Marta failed the exam.
E. Marta passed the exam.

6. A solution is diluted so that its concentration falls by half every hour. After 4 hours, what fraction of the original concentration remains?

A. 1/16
B. 1/8
C. 1/12
D. 1/32
E. 1/4

7. Anna is older than Bruno. Carla is younger than Bruno but older than Dario. Elena is older than Anna. Who is the second youngest?

A. Anna
B. Carla
C. Dario
D. Elena
E. Bruno

8. A box has a volume of 240 cm^3. Its length is 10 cm and its width is 6 cm. What is its height?

A. 4 cm
B. 6 cm
C. 2 cm
D. 3 cm
E. 24 cm

9. In a survey, 60% of respondents were women, and 25% of the women smoke. If 300 people were surveyed, how many women smoke?

A. 180
B. 60
C. 45
D. 30
E. 75

Section 3 — Biology (Q10–32)

10. Which statement about the Golgi apparatus is correct?

A. It is found only in plant cells.
B. It produces ATP.
C. It modifies, sorts and packages proteins received from the endoplasmic reticulum.
D. It contains the cell's DNA.
E. It synthesises lipids for the nucleus.

11. In eukaryotic mRNA processing, which of these does NOT normally occur?

A. Export from the nucleus
B. Splicing out of introns
C. Addition of a 5' cap
D. Removal of exons
E. Addition of a poly-A tail

12. Which of the following is a correct comparison of mitosis and meiosis in humans?

A. Both produce four daughter cells.
B. Mitosis produces haploid cells; meiosis produces diploid cells.
C. Mitosis produces genetically identical cells; meiosis produces genetically varied haploid cells.
D. Both involve two nuclear divisions.
E. Crossing over is typical of mitosis.

13. In which location does the Krebs cycle occur in eukaryotic cells?

A. Intermembrane space
B. Mitochondrial matrix
C. Inner mitochondrial membrane
D. Cytoplasm
E. Nucleus

14. Haemophilia A is X-linked recessive. An affected man has children with a homozygous normal woman. Which is true?

A. Half of the daughters will be affected.
B. All sons will be affected.
C. All children will be unaffected and non-carriers.
D. All daughters will be carriers.
E. Half of the sons will be carriers.

15. Which of the following is a product of the light-dependent reactions of photosynthesis?

A. Pyruvate
B. Ribulose bisphosphate
C. CO_2
D. Glucose
E. ATP and NADPH

16. The main function of red blood cells is to:
A. produce antibodies.
B. digest pathogens.
C. clot blood.
D. fight infection.
E. transport oxygen using haemoglobin.

17. Which of the following molecules is NOT a lipid?
A. Testosterone
B. Phospholipid
C. Cholesterol
D. Glycogen
E. Triglyceride

18. The myelin sheath in the peripheral nervous system is formed by:
A. Oligodendrocytes
B. Microglia
C. Neurons themselves
D. Schwann cells
E. Astrocytes

19. Which valve prevents backflow of blood from the left ventricle into the left atrium?
A. Tricuspid valve
B. Aortic valve
C. Mitral (bicuspid) valve
D. Pulmonary valve
E. Semilunar valve of the vena cava

20. In snapdragons, flower colour shows incomplete dominance: RR is red, rr is white, Rr is pink. Crossing two pink plants gives which phenotypic ratio?
A. 1 red : 2 pink : 1 white
B. 3 red : 1 white
C. All pink
D. 9 : 3 : 3 : 1
E. 1 red : 1 white

21. Thyroxine production depends on adequate dietary intake of:
A. Calcium
B. Iron
C. Zinc
D. Sodium
E. Iodine

22. The DNA of a species contains 30% adenine. What percentage is guanine?
A. 60%
B. 20%
C. 30%
D. 70%
E. 40%

23. Which cells directly kill virus-infected body cells?

A. Mast cells
B. Erythrocytes
C. Cytotoxic ($CD8^+$) T cells
D. Helper T cells
E. Plasma cells

24. In the loop of Henle, which part is permeable to water but largely impermeable to solutes?

A. Descending limb
B. Glomerulus
C. Thin ascending limb
D. Macula densa
E. Thick ascending limb

25. A diploid organism has 2n = 8. How many different combinations of chromosomes can be produced in gametes by independent assortment alone?

A. 4
B. 256
C. 8
D. 16
E. 64

26. Which bond links amino acids in a polypeptide?

A. Phosphodiester
B. Peptide
C. Ester
D. Hydrogen
E. Glycosidic

27. Which of the following is an example of a chromosomal aneuploidy?

A. Huntington's disease
B. Sickle cell anaemia
C. Cystic fibrosis
D. Down syndrome (trisomy 21)
E. Haemophilia

28. A red blood cell is placed in a hypotonic solution. What happens?

A. It actively pumps water out and stays the same.
B. It divides.
C. Nothing happens.
D. It shrinks (crenates).
E. It swells and may burst (haemolysis).

29. Which structure is used by many bacteria for movement?

A. Nucleoid
B. Flagellum
C. Plasmid
D. Pili
E. Capsule

30. Pepsin is an enzyme that digests:
A. carbohydrates in the mouth.
B. nucleic acids in the colon.
C. starch in the stomach.
D. proteins in the stomach.
E. fats in the small intestine.

31. During translation, the ribosome moves along the mRNA in which direction?
A. 5' → 3'
B. N-terminus to C-terminus of the mRNA
C. 3' → 5'
D. Both directions alternately
E. It does not move

32. Which of the following best describes homologous structures?
A. Structures produced by convergent evolution.
B. Structures that have lost their function.
C. Structures found only in plants.
D. Structures with common evolutionary origin, possibly different functions.
E. Structures with similar function but different evolutionary origin.

Section 4 — Chemistry (Q33–47)

33. How many molecules are in 0.25 mol of CO_2? ($N_A = 6 \times 10^{23}$ mol^{-1})
A. 2.4×10^{24}
B. 6×10^{23}
C. 1.5×10^{24}
D. 2.4×10^{23}
E. 1.5×10^{23}

34. In the reaction $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$, which species is oxidised?
A. No species is oxidised.
B. I_2
C. Fe^{3+}
D. Fe^{2+}
E. I^-

35. A solution has $[OH^-] = 1 \times 10^{-5}$ mol/L at 25 °C. What is its pH?
A. 9
B. 7
C. 10
D. 14
E. 5

36. Which of the following has the largest ionic radius?
A. Na^+
B. O^{2-}
C. F^-
D. Mg^{2+}
E. Al^{3+}

37. What is the hybridisation of carbon in methane, CH_4?

A. sp^3
B. sp^3d
C. sp^2
D. sp
E. Unhybridised

38. 10 g of $CaCO_3$ (M = 100 g/mol) is heated: $CaCO_3 \rightarrow CaO + CO_2$. What mass of CO_2 (M = 44 g/mol) is produced?

A. 5.6 g
B. 4.4 g
C. 2.2 g
D. 10 g
E. 44 g

39. Which reaction converts an alkene into an alkane?

A. Hydration
B. Hydrogenation
C. Halogenation with Br_2
D. Oxidation
E. Esterification

40. How many grams of glucose ($C_6H_{12}O_6$, M = 180 g/mol) are needed to make 250 mL of a 0.2 mol/L solution?

A. 90 g
B. 0.9 g
C. 9 g
D. 36 g
E. 3.6 g

41. Which substance has a giant covalent structure?

A. Copper
B. NaCl
C. Diamond
D. Iodine
E. Methane

42. Which of the following is a strong acid in water?

A. H_2CO_3
B. HNO_3
C. CH_3COOH
D. NH_4^+
E. HF

43. A catalyst affects a reaction by:

A. providing an alternative pathway with lower activation energy.
B. being permanently consumed.
C. increasing the yield at equilibrium.
D. changing ΔH of the reaction.
E. shifting equilibrium to the right.

44. What is the IUPAC name of $CH_3CH_2CH_2OH$?

A. Ethanol
B. Propanal
C. Propanone
D. Propan-1-ol
E. Propan-2-ol

45. How many electrons can the n = 3 shell hold at most?

A. 2
B. 8
C. 18
D. 32
E. 10

46. Which statement about a galvanic (voltaic) cell is correct?

A. It converts electrical energy into chemical energy.
B. Electrons flow from cathode to anode in the external circuit.
C. Reduction takes place at the cathode.
D. It requires an external power source.
E. The anode is always positive.

47. Which molecule is chiral?

A. CHBrClF
B. CH_2Cl_2
C. CH_3Cl
D. CCl_4
E. CH_3CH_2OH

Section 5 — Physics & Mathematics (Q48–60)

48. A ball is thrown vertically upward at 20 m/s (g = 10 m/s²). What maximum height does it reach?

A. 20 m
B. 200 m
C. 80 m
D. 10 m
E. 40 m

49. A 60 kg person climbs 5 m of stairs in 10 s (g = 10 m/s²). What average power is developed?

A. 30 W
B. 3000 W
C. 600 W
D. 60 W
E. 300 W

50. A 12 V battery is connected across a 4 Ω resistor. What power is dissipated?
A. 3 W
B. 48 W
C. 36 W
D. 12 W
E. 144 W

51. A sound wave travels at 340 m/s and has a wavelength of 0.5 m. What is its frequency?
A. 680 Hz
B. 1360 Hz
C. 0.0015 Hz
D. 340 Hz
E. 170 Hz

52. A gas is heated at constant pressure from 300 K to 600 K. Its volume:
A. quadruples.
B. stays the same.
C. increases by 300 times.
D. halves.
E. doubles.

53. A 2 kg ball moving at 6 m/s collides with and sticks to a stationary 4 kg ball. What is their common speed after the collision?
A. 2 m/s
B. 6 m/s
C. 1 m/s
D. 3 m/s
E. 4 m/s

54. How much heat is needed to raise the temperature of 2 kg of water by 10 °C? (c = 4200 J/(kg·°C))
A. 8400 J
B. 840 J
C. 84 000 J
D. 840 000 J
E. 42 000 J

55. What is $\log_{10}(0.001)$?
A. 2
B. −2
C. 0.001
D. −3
E. 3

56. What is cos(120°)?
A. √3/2
B. 1/2
C. 0
D. −√3/2
E. −1/2

57. A fair coin is tossed 3 times. What is the probability of getting exactly 2 heads?

A. 2/3
B. 1/2
C. 1/4
D. 1/8
E. 3/8

58. What is the distance between the points $(-1, 2)$ and $(2, 6)$?

A. 4
B. 5
C. 7
D. 25
E. 3

59. If $f(x) = 2x^2 - 3x + 1$, what is $f(-1)$?

A. 0
B. 2
C. -4
D. 6
E. 4

60. Two point charges are separated by distance r. If r is doubled, the electrostatic force between them becomes:

A. half as large.
B. 4 times larger.
C. 2 times larger.
D. one quarter as large.
E. unchanged.

정답 및 해설

번호	정답	해설
1	E	꿈이 기억 강화에 기여하는지는 "여전히 논쟁 중".
2	B	학습–시험 사이 경과 시간을 동일하게 통제.
3	B	천연두(1980년 근절 선언).
4	C	단테 알리기에리.
5	E	"오직 합격자만 등록 가능" → 등록했으면 합격.
6	A	$(1/2)^4 = 1/16$.
7	B	나이순: E > A > B > C > D → 두 번째로 어린 사람 Carla.
8	A	$240/(10\times6) = 4$.
9	C	$300\times0.6 = 180$, $\times0.25 = 45$.
10	C	골지체: 단백질 수식·분류·포장.
11	D	제거되는 것은 인트론, 엑손은 남음.
12	C	체세포분열: 동일 유전자 / 감수분열: 다양한 반수체.
13	B	미토콘드리아 기질.
14	D	아버지 X^h → 딸 모두 보인자, 아들은 Y 받아 정상.
15	E	명반응 산물: ATP, NADPH (+O_2).
16	E	헤모글로빈으로 O_2 운반.
17	D	글리코겐은 탄수화물.
18	D	말초 = 슈반세포, 중추 = 희소돌기아교세포.
19	C	승모판.
20	A	불완전 우성 1:2:1.
21	E	요오드.
22	B	A=T=30 → G+C=40 → G=20%.
23	C	세포독성 T세포.
24	A	하행지: 물 투과, 용질 거의 불투과.
25	D	$n=4 \rightarrow 2^4 = 16$.
26	B	펩타이드 결합.
27	D	다운 증후군(21번 삼염색체).
28	E	저장액 → 물 유입 → 용혈.
29	B	편모.
30	D	펩신: 위에서 단백질 분해.
31	A	리보솜은 mRNA를 5'→3'으로 읽음.
32	D	상동기관 = 공통 기원, 기능은 다를 수 있음.
33	E	$0.25 \times 6\times10^{23} = 1.5\times10^{23}$.
34	E	$I^-(-1) \rightarrow I_2(0)$: 산화.
35	A	pOH 5 → pH 9.

번호	정답	해설
36	B	등전자(10e⁻) 중 양성자 수 가장 적은 O^{2-}가 가장 큼.
37	A	sp^3.
38	B	0.1 mol → CO_2 0.1 mol = 4.4 g.
39	B	수소화(H_2, Ni/Pt 촉매).
40	C	0.2×0.25 = 0.05 mol × 180 = 9 g.
41	C	다이아몬드(거대 공유결합 구조).
42	B	HNO_3 강산.
43	A	활성화에너지가 낮은 대체 경로.
44	D	프로판-1-올.
45	C	$2n^2 = 18$.
46	C	환원은 cathode. 갈바니 전지에서 전자는 anode→cathode.
47	A	네 치환기가 모두 다른 탄소.
48	A	$h = v^2/2g = 400/20 = 20$ m.
49	E	$mgh/t = 60×10×5/10 = 300$ W.
50	C	$V^2/R = 144/4 = 36$ W.
51	A	$f = v/\lambda = 680$ Hz.
52	E	샤를 법칙 $V \propto T$.
53	A	$12 = 6v \rightarrow v = 2$ m/s.
54	C	$Q = mc\Delta T = 2×4200×10 = 84\,000$ J.
55	D	10^{-3}.
56	E	$\cos 120° = -1/2$.
57	E	${}_3C_2/8 = 3/8$.
58	B	$\sqrt{(9+16)} = 5$.
59	D	2+3+1 = 6.
60	D	쿨롱 법칙 $F \propto 1/r^2$.

IMAT 모의고사 4회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Urban heat islands form because asphalt, concrete and dark roofs absorb sunlight during the day and release heat slowly at night, while cities have fewer trees to provide shade and evaporative cooling. During heatwaves, night-time temperatures in dense neighbourhoods can remain several degrees higher than in nearby countryside, and hospital admissions among older residents rise sharply. Planting street trees and using reflective roofing are among the cheapest interventions, but their benefits are unevenly distributed: wealthier districts tend to have more green space already.

1. Which of the following is given in the passage as a reason why cities stay warmer at night?

A. Older residents use more heating.
B. Building materials release absorbed heat slowly.
C. Hospitals generate heat.
D. Cities receive more sunlight than the countryside.
E. Reflective roofs trap heat.

2. What concern does the author raise in the final sentence?

A. The benefit of greening may not reach the neighbourhoods that need it most.
B. Reflective roofs are too expensive.
C. Trees increase night-time temperatures.
D. Wealthy districts suffer most from heatwaves.
E. Heat islands do not affect health.

3. Which organ was first successfully transplanted between humans (identical twins) in 1954?

A. Heart
B. Cornea
C. Kidney
D. Lung
E. Liver

4. In which year did the Berlin Wall fall?

A. 1975
B. 1985
C. 1961
D. 1991
E. 1989

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "No reptiles are mammals. All snakes are reptiles." Which conclusion is valid?

A. No mammals are reptiles and some are snakes.
B. All reptiles are snakes.
C. Some snakes are mammals.
D. Some mammals are snakes.
E. No snakes are mammals.

6. A nurse can prepare 12 infusion bags per hour; a second nurse can prepare 8 per hour. Working together, how long do they take to prepare 50 bags?

A. 2 h
B. 4 h 10 min
C. 4 h
D. 3 h
E. 2 h 30 min

7. A code replaces each letter with the letter two places later in the alphabet (A→C, B→D, ..., Y→A, Z→B). How is the word "ZEBRA" written?

A. XCZPY
B. BGDTA
C. AFCSB
D. BGDTC
E. BGDSC

8. The average of five numbers is 12. When a sixth number is added, the average becomes 13. What is the sixth number?

A. 13
B. 18
C. 20
D. 15
E. 25

9. A clinical trial has 240 participants divided between drug and placebo in the ratio 5 : 3. How many receive the placebo?

A. 80
B. 90
C. 48
D. 120
E. 150

Section 3 — Biology (Q10–32)

10. Which organelle contains hydrolytic enzymes that digest worn-out organelles?

A. Lysosome
B. Vacuole of animal cells
C. Centriole
D. Nucleolus
E. Ribosome

11. Which enzyme synthesises an RNA primer during DNA replication?
A. Reverse transcriptase
B. Primase
C. Ligase
D. Helicase
E. DNA polymerase I

12. Which checkpoint protein is often called "the guardian of the genome" and can halt the cell cycle when DNA is damaged?
A. Tubulin
B. Insulin receptor
C. p53
D. Cyclin B
E. Actin

13. In the absence of oxygen, human muscle cells convert pyruvate to:
A. ethanol and CO_2.
B. acetyl-CoA.
C. citrate.
D. lactate.
E. glucose.

14. In a pedigree, two unaffected parents have an affected daughter. What is the most likely mode of inheritance?
A. X-linked dominant
B. X-linked recessive
C. Y-linked
D. Autosomal dominant
E. Autosomal recessive

15. Which pigment absorbs light energy for photosynthesis in plants?
A. Haemoglobin
B. Chlorophyll
C. Melanin
D. Myoglobin
E. Carotene only

16. Gas exchange in the lungs takes place in the:
A. alveoli.
B. bronchioles.
C. trachea.
D. bronchi.
E. pleura.

17. Which of the following is a monosaccharide?
A. Amylose
B. Fructose
C. Maltose
D. Lactose
E. Sucrose

18. At rest, the membrane potential of a typical neuron is approximately:
A. +30 mV
B. −150 mV
C. +70 mV
D. −70 mV
E. 0 mV

19. The pacemaker of the heart is the:
A. aortic valve.
B. bundle of His.
C. atrioventricular node.
D. sinoatrial node.
E. Purkinje fibres.

20. In a cross Aa × aa, what proportion of offspring will be homozygous?
A. 1/4
B. 3/4
C. 1/2
D. 1
E. 0

21. Which gland secretes adrenaline?
A. Adrenal medulla
B. Pancreas
C. Thyroid
D. Pituitary
E. Adrenal cortex

22. Nucleotides are made of:
A. a fatty acid and glycerol.
B. an amino acid, a sugar and a phosphate.
C. a nitrogenous base, a pentose sugar and a phosphate group.
D. glucose and a phosphate group.
E. a nitrogenous base and an amino acid.

23. Which immunoglobulin class can cross the placenta?
A. IgA
B. IgE
C. IgD
D. IgG
E. IgM

24. Antidiuretic hormone (ADH) acts by:
A. increasing glomerular filtration rate.
B. stimulating renin release.
C. reducing water reabsorption in the collecting duct.
D. stimulating sodium excretion.
E. increasing water permeability of the collecting duct.

25. A cell has 24 chromosomes at the end of telophase II of meiosis. How many chromosomes did the parent cell have before meiosis began?

A. 48
B. 12
C. 72
D. 24
E. 96

26. Which of the following stabilises the α-helix of a protein's secondary structure?

A. Phosphodiester bonds
B. Disulfide bridges between distant regions
C. Hydrogen bonds between backbone C=O and N–H groups
D. Glycosidic bonds
E. Ionic bonds between side chains only

27. Sickle cell anaemia is caused by:

A. a single base substitution in the β-globin gene.
B. a frameshift mutation in insulin.
C. a deletion of a whole chromosome.
D. trisomy.
E. an X-linked dominant gene.

28. Which statement about osmosis is correct?

A. Water moves across a partially permeable membrane from a region of lower solute concentration to higher solute concentration.
B. Osmosis is the movement of solutes.
C. Water moves from a region of higher solute concentration to lower solute concentration.
D. Osmosis only occurs in plants.
E. Osmosis requires ATP.

29. Which method of horizontal gene transfer in bacteria requires direct cell-to-cell contact via a pilus?

A. Binary fission
B. Conjugation
C. Transduction
D. Transformation
E. Sporulation

30. Which enzyme, secreted in saliva, begins starch digestion?

A. Lactase
B. Trypsin
C. Amylase
D. Lipase
E. Pepsin

31. A polypeptide of 150 amino acids is coded for by a minimum of how many nucleotides of mRNA (including the stop codon)?

A. 300
B. 450
C. 600
D. 150
E. 453

32. A species that has evolved into several species, each adapted to a different niche, is an example of:

A. convergent evolution.
B. genetic drift.
C. co-dominance.
D. adaptive radiation.
E. extinction.

Section 4 — Chemistry (Q33–47)

33. What mass of NaCl (M = 58.5 g/mol) contains 2 mol of chloride ions?

A. 117 g
B. 234 g
C. 29.25 g
D. 87.75 g
E. 58.5 g

34. What is the oxidation number of sulfur in H_2SO_3?

A. +6
B. −2
C. +8
D. +4
E. +2

35. Mixing equal volumes of 0.2 mol/L HCl and 0.1 mol/L NaOH gives a solution with $[H^+]$ of:

A. 0.3 mol/L
B. 0.15 mol/L
C. 0.05 mol/L
D. 0.01 mol/L
E. 0.1 mol/L

36. Electronegativity generally:

A. is the same for all halogens.
B. increases across a period from left to right.
C. is highest for alkali metals.
D. decreases across a period from left to right.
E. increases down a group.

37. How many σ and π bonds are in a molecule of ethene, C_2H_4?

A. 6 σ, 0 π
B. 4 σ, 2 π
C. 5 σ, 1 π
D. 5 σ, 2 π
E. 4 σ, 1 π

38. In the reaction $2Al + 3Cl_2 \rightarrow 2AlCl_3$, how many moles of Cl_2 are needed to react with 6 mol of Al?

A. 12
B. 6
C. 9
D. 4
E. 3

39. Which reaction produces an ester?

A. Alcohol + oxygen
B. Alkane + chlorine
C. Alcohol + carboxylic acid (acid catalyst)
D. Aldehyde + hydrogen
E. Alkene + water

40. A solution contains 4.9 g of H_2SO_4 (M = 98 g/mol) in 250 mL. What is its molar concentration?

A. 0.1 mol/L
B. 0.4 mol/L
C. 0.2 mol/L
D. 0.05 mol/L
E. 2 mol/L

41. Which compound has the highest boiling point?

A. CH_4
B. All have the same boiling point
C. C_3H_8
D. C_4H_{10} (n-butane)
E. C_2H_6

42. A buffer is made from NH_3 and NH_4Cl. If a small amount of HCl is added, which species reacts with it?

A. NH_3
B. NH_4^+
C. H_2O
D. Cl^-
E. OH^- produced by NH_4^+

43. For a reaction at equilibrium, $K_c = 10^{-5}$. This indicates that:

A. the reaction is exothermic.
B. the reaction cannot occur.
C. the products are strongly favoured.
D. the reactants are favoured at equilibrium.
E. the reaction is very fast.

44. Which of these compounds is an amine?

A. CH_3CN
B. CH_3NO_2
C. CH_3COOH
D. CH_3NH_2
E. CH_3CONH_2

45. An element has electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$. To which group does it belong?

A. Group 1
B. Group 2
C. Group 17
D. Group 15
E. Group 16

46. In the reaction $2H_2O_2 \rightarrow 2H_2O + O_2$, the oxygen in H_2O_2 is:

A. both oxidised and reduced (disproportionation).
B. neither oxidised nor reduced.
C. only reduced.
D. converted to ozone.
E. only oxidised.

47. Benzene is best described as:

A. an ionic compound.
B. a saturated hydrocarbon.
C. a cycloalkane with three isolated double bonds.
D. a linear alkyne.
E. an aromatic ring with delocalised π electrons.

Section 5 — Physics & Mathematics (Q48–60)

48. A cyclist travels 120 m in 8 s at constant speed, then 80 m in 12 s. What is the average speed for the whole trip?

A. 12 m/s
B. 10 m/s
C. 20 m/s
D. 15 m/s
E. 8 m/s

49. A spring with spring constant 200 N/m is stretched by 0.1 m. How much elastic potential energy is stored?

A. 20 J
B. 10 J
C. 0.5 J
D. 2 J
E. 1 J

50. A 4 Ω resistor and a 12 Ω resistor in parallel are connected in series with a 2 Ω resistor. What is the total resistance?

A. 14 Ω
B. 5 Ω
C. 18 Ω
D. 3 Ω
E. 8 Ω

51. If the frequency of a wave is doubled while its speed remains constant, its wavelength:

A. halves.
B. doubles.
C. becomes zero.
D. stays the same.
E. quadruples.

52. A hydraulic press has a small piston of area 0.01 m² and a large piston of area 0.5 m². A force of 100 N is applied to the small piston. What force is exerted by the large piston?

A. 2 N
B. 50 000 N
C. 500 N
D. 50 N
E. 5000 N

53. An object moves in a circle of radius 2 m at a constant speed of 4 m/s. What is its centripetal acceleration?

A. 2 m/s²
B. 32 m/s²
C. 4 m/s²
D. 16 m/s²
E. 8 m/s²

54. A lamp rated 60 W is used for 5 hours. How much energy does it use?

A. 3 kWh
B. 300 kWh
C. 12 kWh
D. 0.3 kWh
E. 0.012 kWh

55. Solve: 2^(x+1) = 32.

A. 16
B. 5
C. 6
D. 3
E. 4

56. If sin θ = 3/5 and θ is acute, what is tan θ?

A. 4/5
B. 3/4
C. 3/5
D. 4/3
E. 5/3

57. How many different ways can 4 different books be arranged on a shelf?

A. 4
B. 16
C. 8
D. 12
E. 24

58. Which of the following is the equation of a line parallel to $y = 3x + 2$ passing through $(0, -1)$?

A. $y = -3x - 1$
B. $y = -(1/3)x + 2$
C. $y = (1/3)x - 1$
D. $y = 3x - 1$
E. $y = 3x + 1$

59. Solve the inequality $3 - 2x > 7$.

A. $x < -2$
B. $x < 5$
C. $x > -2$
D. $x < 2$
E. $x > 2$

60. A radioactive sample has a half-life of 6 hours. What fraction remains after 24 hours?

A. 1/4
B. 1/16
C. 1/24
D. 1/8
E. 1/32

정답 및 해설

번호	정답	해설
1	B	아스팔트·콘크리트가 흡수한 열을 천천히 방출.
2	A	녹화 효과가 이미 녹지 많은 부유층 지역에 쏠림 → 형평성 우려.
3	C	1954년 일란성 쌍둥이 간 신장 이식(머리, 보스턴).
4	E	1989년.
5	E	뱀 ⊂ 파충류, 파충류 ∩ 포유류 = ∅ → 뱀은 포유류가 아님.
6	E	20개/h → 50/20 = 2.5 h.
7	D	Z→B, E→G, B→D, R→T, A→C → BGDTC.
8	B	13×6 − 12×5 = 78 − 60 = 18.
9	B	240 × 3/8 = 90.
10	A	리소좀(자가포식).
11	B	프라이메이스.
12	C	p53.
13	D	젖산 발효.
14	E	정상 부모 → 이환 딸: 상염색체 열성(X연관 열성이면 아버지가 이환이어야 함).
15	B	엽록소.
16	A	폐포.
17	B	과당.
18	D	약 −70 mV.
19	D	동방결절(SA node).
20	C	Aa × aa → Aa 1/2(이형), aa 1/2(동형) → 동형접합 1/2.
21	A	부신수질.
22	C	염기 + 5탄당 + 인산.
23	D	IgG.
24	E	집합관에 아쿠아포린 삽입 → 물 투과성↑.
25	A	감수분열 산물 n=24 → 모세포 2n=48.
26	C	주사슬 C=O와 N–H 사이 수소결합.
27	A	β-글로빈 유전자 단일 염기 치환(Glu→Val).
28	A	저농도 용액 → 고농도 용액 쪽으로 물 이동.
29	B	접합(conjugation).
30	C	침샘 아밀레이스.
31	E	150×3 + 종결코돈 3 = 453.
32	D	적응방산.
33	A	2 mol × 58.5 = 117 g.
34	D	2(+1) + S + 3(−2) = 0 → +4.
35	C	남는 H^+ 0.1 mol/L·V, 부피 2V → 0.05 mol/L.

번호	정답	해설
36	B	주기에서 오른쪽으로 갈수록 증가.
37	C	C–H 4 + C–C σ 1 = 5σ, π 1.
38	C	2:3 → 6 mol Al : 9 mol Cl_2.
39	C	에스터화(피셔 에스터화).
40	C	0.05 mol / 0.25 L = 0.2 mol/L.
41	D	분자량 클수록 분산력↑.
42	A	$NH_3 + H^+ \rightarrow NH_4^+$.
43	D	$K \ll 1$ → 반응물 쪽 우세.
44	D	메틸아민. E는 아마이드.
45	C	최외각 7개 → 17족(Cl).
46	A	O(−1) → H_2O(−2) 환원, O_2(0) 산화 → 불균등화.
47	E	비편재화된 π 전자를 가진 방향족 고리.
48	B	200 m / 20 s = 10 m/s.
49	E	$\frac{1}{2}kx^2 = \frac{1}{2}\times200\times0.01 = 1$ J.
50	B	4∥12 = 3 Ω, +2 = 5 Ω.
51	A	$\lambda = v/f$.
52	E	100 × 0.5/0.01 = 5000 N.
53	E	$v^2/r = 16/2 = 8$.
54	D	0.06 kW × 5 h = 0.3 kWh.
55	E	$2^5 \rightarrow x+1 = 5$.
56	B	cos = 4/5 → tan = 3/4.
57	E	4! = 24.
58	D	기울기 3, 절편 −1.
59	A	$-2x > 4 \rightarrow x < -2$.
60	B	4번 반감 → 1/16.

IMAT 모의고사 5회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Placebo effects are often dismissed as "imaginary", but brain-imaging studies show that expecting pain relief can trigger the release of the body's own opioids. Remarkably, some patients improve even when told openly that they are receiving a placebo, although such "open-label" trials have mostly involved conditions with subjective symptoms, such as chronic back pain or irritable bowel syndrome. Researchers caution that a placebo cannot shrink a tumour or cure an infection, and that relying on it in place of effective treatment would be unethical.

1. The author mentions brain-imaging studies primarily in order to:

A. argue that pain is imaginary.
B. criticise open-label trials.
C. show that placebo responses have a measurable biological basis.
D. prove that placebos can cure infections.
E. recommend placebos instead of drugs.

2. Which of the following is an assumption underlying the researchers' caution in the last sentence?

A. Open-label placebos are illegal.
B. Effective treatments exist for some conditions that placebos cannot treat.
C. Tumours respond only to placebos.
D. Placebos never have any effect.
E. All symptoms are subjective.

3. Which scientist discovered penicillin in 1928?

A. Jonas Salk
B. Selman Waksman
C. Paul Ehrlich
D. Joseph Lister
E. Alexander Fleming

4. Which of the following countries is NOT a member of the European Union (as of 2026)?

A. Portugal
B. Norway
C. Austria
D. Ireland
E. Croatia

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "Every time it rains, the clinic car park floods. The car park is flooded today." Which of the following is true?

A. The flooding may have another cause; we cannot be sure it rained.
B. Rain never causes flooding.
C. It did not rain today.
D. It must have rained today.
E. It will rain tomorrow.

6. A patient's weight dropped from 80 kg to 68 kg. By what percentage did it decrease?

A. 12%
B. 17.6%
C. 18%
D. 15%
E. 20%

7. Four friends — Luca, Maria, Nina and Omar — each study a different subject: anatomy, biology, chemistry, physics. Luca does not study biology or physics. Maria studies chemistry. Nina does not study physics. What does Omar study?

A. Physics
B. Chemistry
C. Anatomy
D. Biology
E. Cannot be determined

8. A cube of side 3 cm is painted on all faces and then cut into 1 cm cubes. How many small cubes have exactly two painted faces?

A. 18
B. 6
C. 8
D. 27
E. 12

9. A dose of 5 mg/kg of a drug is prescribed. The drug is available as a solution of 25 mg/mL. What volume should be given to a child weighing 20 kg?

A. 0.8 mL
B. 5 mL
C. 2 mL
D. 4 mL
E. 100 mL

Section 3 — Biology (Q10–32)

10. Which of the following best explains why mitochondria are thought to have evolved from free-living bacteria?

A. They carry out photosynthesis.
B. They contain cellulose.
C. They lack membranes.
D. They are found in all prokaryotes.
E. They have their own circular DNA and 70S-like ribosomes, and a double membrane.

11. A DNA molecule replicates twice in a medium containing only ^{14}N, having originally been fully labelled with ^{15}N. What fraction of the resulting DNA molecules contain any ^{15}N?

A. 1/2
B. 0
C. 3/4
D. 1
E. 1/4

12. Which molecule regulates progression through the cell cycle by binding to cyclin-dependent kinases?

A. Tubulin
B. Histones
C. Cyclins
D. Collagen
E. Keratin

13. Per molecule of glucose, how many molecules of CO_2 are released in the complete aerobic oxidation (link reaction + Krebs cycle)?

A. 4
B. 6
C. 3
D. 12
E. 2

14. In humans, brown eyes (B) are dominant to blue eyes (b). Two brown-eyed parents have a blue-eyed child. What is the probability that their next child has brown eyes?

A. 1/2
B. 3/4
C. 2/3
D. 1
E. 1/4

15. What is the main role of the enzyme RuBisCO?

A. Splitting water in photosystem II
B. Synthesising ATP
C. Reducing $NADP^+$
D. Fixing CO_2 to ribulose bisphosphate
E. Producing oxygen

16. Which of the following increases the affinity of haemoglobin for oxygen (shifts the dissociation curve to the left)?

A. Increased CO_2
B. Decreased pH
C. Decreased temperature
D. Increased 2,3-BPG
E. Increased temperature

17. Which of the following is an unsaturated fatty acid characteristic?

A. One or more C=C double bonds
B. Contains no carboxyl group
C. Only single C–C bonds
D. Made of glucose units
E. Solid at room temperature in all cases

18. Which neurotransmitter is released at the neuromuscular junction of skeletal muscle?

A. Noradrenaline
B. Serotonin
C. GABA
D. Dopamine
E. Acetylcholine

19. Cardiac output equals:

A. blood pressure × heart rate.
B. heart rate + stroke volume.
C. heart rate × stroke volume.
D. end-diastolic volume − heart rate.
E. stroke volume / heart rate.

20. Genes A and B are linked. A test cross of AaBb (AB/ab) × aabb gives: AB 42%, ab 42%, Ab 8%, aB 8%. What is the recombination frequency?

A. 8%
B. 84%
C. 50%
D. 16%
E. 42%

21. Insulin lowers blood glucose mainly by:

A. stimulating gluconeogenesis.
B. stimulating glycogen breakdown.
C. increasing glucose uptake into muscle and adipose cells.
D. inhibiting glucose uptake into cells.
E. increasing glucagon secretion.

22. Which statement about DNA is correct?

A. A–T pairs have three hydrogen bonds.
B. The sugar is ribose.
C. Purines pair with purines.
D. The two strands run parallel.
E. G–C pairs have three hydrogen bonds.

23. Which of the following is part of the innate (non-specific) immune response?

A. Phagocytosis by neutrophils
B. Vaccination
C. Antibody production by plasma cells
D. Memory B cells
E. Clonal selection of T cells

24. Which substance is normally found in the glomerular filtrate?

A. Platelets
B. Large plasma proteins such as albumin in large amounts
C. Glucose
D. White blood cells
E. Red blood cells

25. Non-disjunction during meiosis I in a human produces gametes with chromosome numbers of:

A. 21 and 25
B. 23 and 23
C. 23 and 46
D. 22 and 24
E. 46 and 46

26. Denaturation of a protein by heat disrupts mainly which levels of structure?

A. Primary only
B. Peptide bonds
C. Secondary and tertiary
D. None
E. The amino acid sequence

27. Which of the following genetic disorders is inherited as autosomal dominant?

A. Huntington's disease
B. Haemophilia A
C. Phenylketonuria
D. Cystic fibrosis
E. Duchenne muscular dystrophy

28. The Na^+/K^+-ATPase pump moves:

A. 3 Na^+ out and 2 K^+ in per ATP.
B. 2 Na^+ out and 3 K^+ in per ATP.
C. 3 Na^+ in and 2 K^+ out per ATP.
D. Na^+ and K^+ in the same direction.
E. ions without using ATP.

29. Which antibiotic mechanism targets a structure absent in human cells?

A. Inhibition of the Golgi apparatus
B. Inhibition of the nucleus
C. Inhibition of peptidoglycan cell wall synthesis
D. Destruction of cholesterol in membranes
E. Inhibition of mitochondria

30. Which vitamin is absorbed in the terminal ileum and requires intrinsic factor?
A. Vitamin C
B. Vitamin A
C. Vitamin K
D. Vitamin B12
E. Vitamin D

31. The anticodon on a tRNA is 3'-UAC-5'. Which mRNA codon does it pair with?
A. 5'-GUA-3'
B. 5'-CAU-3'
C. 5'-ATG-3'
D. 5'-AUG-3'
E. 5'-UAC-3'

32. Genetic drift has the greatest effect on allele frequencies in:
A. small populations.
B. populations in Hardy–Weinberg equilibrium.
C. asexually reproducing bacteria only.
D. very large populations.
E. populations with high gene flow.

Section 4 — Chemistry (Q33–47)

33. What is the empirical formula of a compound with 40% C, 6.7% H and 53.3% O by mass? (C = 12, H = 1, O = 16)
A. CH_2O
B. C_2H_4O
C. CH_4O
D. CHO
E. $C_6H_{12}O_6$

34. Balance in acidic solution: $MnO_4^- + Fe^{2+} \rightarrow Mn^{2+} + Fe^{3+}$. How many Fe^{2+} ions react with one MnO_4^-?
A. 8
B. 5
C. 3
D. 1
E. 2

35. What is the pH of a solution formed by diluting 10 mL of 0.1 mol/L HCl to 1 L with water?
A. 5
B. 1
C. 2
D. 3
E. 4

36. Which element has the electron configuration [Ar] $4s^2$ $3d^6$?

A. Ni
B. Fe
C. Co
D. Mn
E. Cr

37. Which molecule has a bond angle of about 104.5°?

A. CH_4
B. BF_3
C. CO_2
D. H_2O
E. NH_3

38. In the reaction $N_2 + 3H_2 \rightarrow 2NH_3$, 28 g of N_2 reacts with 3 g of H_2. Which is the limiting reagent, and what mass of NH_3 forms? (N = 14, H = 1)

A. H_2; 17 g
B. Neither; 31 g
C. N_2; 17 g
D. H_2; 34 g
E. N_2; 34 g

39. Which of the following is a ketone?

A. Propanone
B. Propanoic acid
C. Propanal
D. Methyl ethanoate
E. Propan-1-ol

40. What is the osmolarity of a 0.15 mol/L NaCl solution (assuming complete dissociation)?

A. 0.60 osmol/L
B. 0.075 osmol/L
C. 0.30 osmol/L
D. 0.15 osmol/L
E. 0.45 osmol/L

41. Which of the following substances conducts electricity when solid?

A. Diamond
B. Ice
C. NaCl
D. Sugar
E. Graphite

42. The Ka of a weak acid HA is 10^{-5}. What is the pH of a buffer containing equal concentrations of HA and A^-?

A. 9
B. 3
C. 7
D. 5
E. 10

43. Which statement about a reaction with $\Delta H < 0$ is correct?

A. It is always spontaneous.
B. It releases heat to the surroundings.
C. It absorbs heat from the surroundings.
D. It never needs activation energy.
E. Its products have more enthalpy than reactants.

44. How many structural isomers are there for C_5H_{12}?

A. 6
B. 5
C. 4
D. 2
E. 3

45. Which of the following pairs are isotopes?

A. ^{40}K and ^{40}Ca
B. H_2O and D_2O molecules
C. Diamond and graphite
D. O_2 and O_3
E. ^{12}C and ^{14}C

46. Given $E°(Cu^{2+}/Cu) = +0.34$ V and $E°(Zn^{2+}/Zn) = -0.76$ V, what is the standard emf of a Zn/Cu cell?

A. 0.34 V
B. −1.10 V
C. 0.76 V
D. 1.10 V
E. 0.42 V

47. Which type of reaction occurs when an alkane reacts with chlorine under UV light?

A. Nucleophilic substitution
B. Free-radical substitution
C. Condensation
D. Elimination
E. Electrophilic addition

Section 5 — Physics & Mathematics (Q48–60)

48. A projectile is launched horizontally from a cliff 45 m high at 10 m/s. How far from the base of the cliff does it land? ($g = 10$ m/s², no air resistance)

A. 10 m
B. 30 m
C. 20 m
D. 45 m
E. 90 m

49. A 5 kg block slides on a horizontal surface with coefficient of kinetic friction 0.2. What is the friction force? ($g = 10\ m/s^2$)

A. 1 N
B. 25 N
C. 5 N
D. 10 N
E. 2 N

50. A 10 µF capacitor is charged to 20 V. What charge is stored?

A. 2 µC
B. 0.5 µC
C. 30 µC
D. 2000 µC
E. 200 µC

51. Light passes from air (n = 1.0) into glass (n = 1.5). Its speed in glass is: ($c = 3 \times 10^8$ m/s)

A. 4.5×10^7 m/s
B. 3×10^8 m/s
C. 2×10^8 m/s
D. 1.5×10^8 m/s
E. 4.5×10^8 m/s

52. Blood flows through an artery of cross-sectional area 2 cm^2 at 30 cm/s. It then flows into a narrowed section of area 0.5 cm^2. What is the speed there?

A. 120 cm/s
B. 240 cm/s
C. 15 cm/s
D. 7.5 cm/s
E. 60 cm/s

53. A 0.5 kg ball hits a wall at 8 m/s and rebounds at 6 m/s. What is the magnitude of the change in momentum?

A. 14 kg·m/s
B. 3 kg·m/s
C. 7 kg·m/s
D. 4 kg·m/s
E. 1 kg·m/s

54. An ideal heat engine operates between 600 K and 300 K. What is its maximum (Carnot) efficiency?

A. 100%
B. 67%
C. 50%
D. 25%
E. 33%

55. If $\log_3(x) = 4$, what is $\log_9(x)$?

A. 1
B. 8
C. 16
D. 4
E. 2

56. What is the period of the function $y = \sin(2x)$?

A. $\pi/2$
B. 4π
C. 1/2
D. 2π
E. π

57. In a group of 10 students, how many different committees of 3 students can be formed?

A. 720
B. 30
C. 60
D. 1000
E. 120

58. What is the centre of the circle $x^2 + y^2 - 4x + 6y - 3 = 0$?

A. $(-2, 3)$
B. $(2, -3)$
C. $(-4, 6)$
D. $(4, -6)$
E. $(2, 3)$

59. What is the sum of the first 10 terms of the arithmetic sequence 3, 7, 11, ...?

A. 400
B. 210
C. 230
D. 190
E. 39

60. A charge of 2 μC is placed in a uniform electric field of 5000 N/C. What force acts on it?

A. 0.1 N
B. 0.01 N
C. 2500 N
D. 10 000 N
E. 0.001 N

정답 및 해설

번호	정답	해설
1	C	영상 연구는 위약 반응에 생물학적 근거(내인성 오피오이드)가 있음을 보여 주기 위함.
2	B	위약으로 치료할 수 없고 효과적 치료가 따로 있는 질환이 있어야 "대체는 비윤리적"이 성립.
3	E	알렉산더 플레밍.
4	B	노르웨이는 EEA 회원국이지만 EU 비회원.
5	A	후건 긍정의 오류 → 비가 왔다고 확정 불가.
6	D	12/80 = 15%.
7	A	Maria=화학, Luca는 생물·물리 아님 → 해부학. Nina는 물리 아님 → 생물. Omar=물리.
8	E	모서리 중간 큐브: 12개 모서리 × 1 = 12.
9	D	5×20 = 100 mg → 100/25 = 4 mL.
10	E	내공생설 근거: 원형 DNA, 70S 유사 리보솜, 이중막.
11	A	2회 복제 후 4분자 중 2분자가 ^{15}N 가닥 포함 → 1/2.
12	C	사이클린-CDK 복합체.
13	B	연결반응 2 + 크렙스 4 = 6 CO_2.
14	B	부모 Bb × Bb → 갈색 3/4.
15	D	CO_2를 RuBP에 고정.
16	C	온도↓ → 친화력↑(좌측 이동). 나머지는 우측 이동.
17	A	C=C 이중결합 존재.
18	E	아세틸콜린.
19	C	CO = HR × SV.
20	D	재조합형 8+8 = 16%.
21	C	근육·지방세포 GLUT4 → 포도당 흡수↑.
22	E	G–C 수소결합 3개. 두 가닥은 역평행.
23	A	호중구 식세포작용(선천면역).
24	C	포도당은 여과됨(이후 재흡수).
25	D	감수 1분열 비분리 → 24와 22 (n+1, n−1).
26	C	2·3차 구조 파괴, 펩타이드 결합(1차)은 유지.
27	A	헌팅턴병.
28	A	3 Na^+ 밖, 2 K^+ 안.
29	C	펩티도글리칸 세포벽(β-락탐 등).
30	D	비타민 B12 + 내인자, 회장 말단.
31	D	안티코돈 3'-UAC-5' ↔ 코돈 5'-AUG-3'.
32	A	작은 집단.
33	A	C 3.33 : H 6.7 : O 3.33 → 1:2:1 → CH_2O.
34	B	Mn +7→+2 ($5e^-$), Fe +2→+3 ($1e^-$) → 5 Fe^{2+}.
35	D	10^{-3} mol in 1 L → pH 3.

번호	정답	해설
36	B	Fe(26).
37	D	H_2O 굽은형 104.5°.
38	A	N_2 1 mol은 H_2 3 mol 필요, H_2는 1.5 mol뿐 → H_2 한정, NH_3 1 mol = 17 g.
39	A	프로파논(아세톤).
40	C	$Na^+ + Cl^- \rightarrow 2 \times 0.15 = 0.30$.
41	E	흑연(비편재 전자).
42	D	헨더슨–하셀바흐: pH = pKa = 5.
43	B	발열 반응. 자발성은 ΔG로 결정.
44	E	n-펜테인, 2-메틸뷰테인, 2,2-다이메틸프로페인.
45	E	양성자 수 같고 중성자 수 다름.
46	D	$0.34 - (-0.76) = 1.10$ V.
47	B	자유 라디칼 치환.
48	B	낙하 시간 $\sqrt{(2 \cdot 45/10)} = 3$ s → $10 \times 3 = 30$ m.
49	D	$\mu mg = 0.2 \times 5 \times 10 = 10$ N.
50	E	$Q = CV = 10\ \mu F \times 20$ V $= 200\ \mu C$.
51	C	$v = c/n = 2 \times 10^8$.
52	A	$A_1v_1 = A_2v_2 \rightarrow 2 \times 30/0.5 = 120$ cm/s.
53	C	방향 반전: $0.5 \times (8+6) = 7$.
54	C	$1 - 300/600 = 50\%$.
55	E	$x = 81 = 9^2 \rightarrow 2$.
56	E	$2\pi/2 = \pi$.
57	E	${}_{10}C_3 = 120$.
58	B	$(x-2)^2 + (y+3)^2 = 16$ → 중심 $(2, -3)$.
59	B	$a=3, d=4$: $10/2 \times (2 \cdot 3 + 9 \cdot 4) = 5 \times 42 = 210$.
60	B	$F = qE = 2 \times 10^{-6} \times 5000 = 0.01$ N.

IMAT 모의고사 6회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> The human gut contains trillions of microbes, collectively called the microbiota. Observational studies have linked differences in gut microbes to obesity, depression and autoimmune disease. Yet association is not causation: illness, diet and medication can all reshape the microbiota, so altered microbes may be a consequence of disease rather than its cause. The strongest evidence of causation comes from experiments in which microbes transferred from sick humans into germ-free mice produce similar symptoms in the animals.

1. According to the passage, why do observational studies alone fail to prove that microbes cause disease?
A. Disease and its treatment may themselves change the microbiota.
B. The microbiota is too small to study.
C. Obesity is not a disease.
D. Observational studies are always fraudulent.
E. Mice do not have gut microbes.

2. The experiments with germ-free mice are considered stronger evidence because they:
A. introduce the microbes first and then observe whether symptoms follow.
B. measure diet only.
C. remove all microbes from humans.
D. are cheaper.
E. use larger samples of humans.

3. Which Italian scientist is known for pioneering work on nerve growth factor, for which she shared the 1986 Nobel Prize in Physiology or Medicine?
A. Fabiola Gianotti
B. Margherita Hack
C. Maria Montessori
D. Samantha Cristoforetti
E. Rita Levi-Montalcini

4. The Universal Declaration of Human Rights was adopted by the UN General Assembly in:
A. 1957
B. 1966
C. 1948
D. 1919
E. 1945

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. Some antibiotics are effective against viruses — this claim is false. Which of the following must therefore be true?

A. Some viruses are antibiotics.
B. Some antibiotics are not effective against bacteria.
C. All viruses resist antibiotics and bacteria.
D. All antibiotics are effective against viruses.
E. No antibiotics are effective against viruses.

6. A shop increases prices by 20%, and then a further 10%. What is the overall percentage increase?

A. 32%
B. 31%
C. 30%
D. 28%
E. 33%

7. In a hospital, 3 doctors can see 36 patients in 4 hours. At the same rate, how many patients can 5 doctors see in 6 hours?

A. 108
B. 90
C. 80
D. 60
E. 72

8. What is the missing number? 1, 1, 2, 3, 5, 8, 13, __

A. 18
B. 20
C. 26
D. 21
E. 24

9. A runner completes laps of a 400 m track at 80 s per lap. A second runner completes laps at 100 s per lap. They start together. After how many seconds will the faster runner first lap (overtake by one full lap) the slower runner?

A. 200 s
B. 800 s
C. 320 s
D. 1000 s
E. 400 s

Section 3 — Biology (Q10–32)

10. Which cytoskeletal component forms the spindle fibres during cell division?

A. Keratin
B. Collagen fibres
C. Microtubules
D. Intermediate filaments
E. Actin filaments

11. Which statement about the genetic code is correct?

A. The code is degenerate: most amino acids are coded by more than one codon.
B. The code differs completely between humans and bacteria.
C. There are 20 possible codons.
D. Each amino acid is coded by exactly one codon.
E. Codons consist of two nucleotides.

12. Apoptosis is:

A. the fusion of gametes.
B. uncontrolled cell death caused by injury.
C. a form of mitosis.
D. programmed cell death.
E. cell division in bacteria.

13. Which molecule enters the Krebs cycle by combining with oxaloacetate?

A. Lactate
B. Pyruvate
C. NADH
D. Glucose
E. Acetyl-CoA

14. A woman carrier of an X-linked recessive disease and an unaffected man have a daughter. What is the probability that the daughter is a carrier?

A. 1/2
B. 3/4
C. 1/4
D. 0
E. 1

15. In photosynthesis, the oxygen released comes from:

A. ATP
B. glucose
C. chlorophyll
D. water
E. CO_2

16. Which part of the brain is primarily responsible for coordination of balance and fine movement?

A. Thalamus
B. Cerebellum
C. Medulla oblongata
D. Cerebrum
E. Hypothalamus

17. Which of the following describes a phospholipid?

A. A chain of glucose monomers
B. Four fused carbon rings
C. A polymer of amino acids
D. Glycerol, two fatty acids and a phosphate group with a hydrophilic head
E. Three fatty acids attached to glycerol

18. Saltatory conduction occurs because action potentials:

A. travel only in unmyelinated axons.
B. travel backwards along the axon.
C. are produced in the dendrites.
D. are conducted by Schwann cells.
E. jump between nodes of Ranvier in myelinated axons.

19. Which blood component is most important for clot formation?

A. Lymphocytes
B. Plasma water
C. Albumin
D. Platelets
E. Erythrocytes

20. A man of blood group A (heterozygous) and a woman of blood group B (heterozygous) have children. What is the probability that a child has blood group O?

A. 0
B. 1/4
C. 1/2
D. 1/8
E. 3/4

21. The hypothalamus controls the anterior pituitary mainly by:

A. nerve impulses directly to the anterior pituitary.
B. secreting insulin.
C. releasing hormones carried in the hypophyseal portal system.
D. producing ADH stored in the anterior pituitary.
E. producing thyroxine.

22. How many hydrogen bonds hold together a DNA fragment with 10 base pairs, of which 4 are G–C?

A. 20
B. 34
C. 24
D. 30
E. 26

23. Which statement about antibodies is correct?

A. All antibodies bind all antigens.
B. Antibodies are produced by red blood cells.
C. Antibodies consist of two heavy and two light chains.
D. Each antibody has four identical polypeptide chains.
E. Antibodies are carbohydrates.

24. The countercurrent multiplier in the kidney allows:

A. increased blood pressure.
B. secretion of glucose.
C. synthesis of urea.
D. filtration of red blood cells.
E. production of concentrated urine.

25. In which phase of meiosis does crossing over occur?

A. Anaphase II
B. Metaphase I
C. Anaphase I
D. Prophase II
E. Prophase I

26. Which enzyme property explains why it works best at a particular pH?

A. pH changes increase activation energy permanently.
B. pH changes the enzyme into a carbohydrate.
C. pH has no effect on enzymes.
D. pH changes alter the ionisation of amino acids in the active site.
E. pH changes alter the primary structure.

27. Which of the following is a sex-chromosome aneuploidy?

A. Turner syndrome (45,X)
B. Cri-du-chat syndrome
C. Patau syndrome
D. Down syndrome
E. Edwards syndrome

28. Which process moves large particles into a cell by engulfing them in a vesicle?

A. Exocytosis
B. Active transport through a carrier
C. Osmosis
D. Facilitated diffusion
E. Phagocytosis

29. Retroviruses such as HIV use which enzyme to convert their RNA into DNA?

A. Ligase
B. DNA polymerase III
C. RNA polymerase II
D. Reverse transcriptase
E. Integrase only

30. The hormone secretin stimulates the pancreas to secrete:
A. insulin.
B. bile.
C. glucagon.
D. pepsinogen.
E. bicarbonate-rich fluid.

31. Which of the following is the correct order in protein synthesis and secretion?
A. Nucleus → lysosome → Golgi → membrane
B. Smooth ER → nucleus → ribosome → membrane
C. Ribosome → mitochondria → Golgi → membrane
D. Ribosome on rough ER → ER lumen → vesicle → Golgi → secretory vesicle → plasma membrane
E. Golgi → ribosome → rough ER → vesicle → membrane

32. Which of the following provides evidence for evolution from molecular biology?
A. Fossil record
B. Geographical distribution of species
C. Vestigial organs
D. Artificial selection
E. Similar DNA and protein sequences among related species

Section 4 — Chemistry (Q33–47)

33. How many moles of atoms in total are present in 1 mol of glucose, $C_6H_{12}O_6$?
A. 12
B. 24
C. 180
D. 18
E. 6

34. What is the oxidation number of nitrogen in NH_4^+?
A. +3
B. −3
C. −4
D. +4
E. +5

35. A solution of pH 4 is diluted 100 times with water. What is the new pH (approximately)?
A. 5
B. 6
C. 4
D. 2
E. 8

36. Which of the following is a noble gas?

A. N
B. Ne
C. F
D. O
E. Na

37. Which molecule has a trigonal planar shape?

A. NH_3
B. PCl_3
C. CH_4
D. BF_3
E. H_2O

38. How many litres of O_2 at STP (22.4 L/mol) are needed to burn 1 mol of CH_4 completely?

A. 67.2 L
B. 44.8 L
C. 89.6 L
D. 22.4 L
E. 11.2 L

39. What is the product of the hydration of ethene (in the presence of an acid catalyst)?

A. Ethanoic acid
B. Ethyl ethanoate
C. Ethanol
D. Ethanal
E. Ethane

40. 100 mL of 0.5 mol/L NaCl is mixed with 400 mL of water. What is the final concentration?

A. 0.2 mol/L
B. 0.05 mol/L
C. 0.1 mol/L
D. 0.125 mol/L
E. 0.25 mol/L

41. Which of the following molecules can form hydrogen bonds with each other?

A. CH_4
B. CH_3OCH_3
C. CH_3OH
D. CO_2
E. CCl_4

42. Which of the following salts gives a basic solution in water?

A. NH_4Cl
B. NaCl
C. KNO_3
D. CH_3COONa
E. Na_2SO_4

43. For the reaction A + B → C, the rate doubles when [A] doubles and quadruples when [B] doubles. What is the overall order?

A. 4
B. 1
C. 0
D. 3
E. 2

44. Which functional group is present in an amide?

A. $–CONH_2$
B. –COOH
C. –OH
D. $–NH_2$ only
E. –CN

45. Which quantum number determines the shape of an orbital?

A. Magnetic (m_l)
B. Mass number
C. Principal (n)
D. Azimuthal / angular momentum (l)
E. Spin (m_s)

46. In the electrolysis of molten NaCl, what is produced at the cathode?

A. Cl_2
B. O_2
C. Na
D. H_2
E. NaOH

47. Which compound is an isomer of ethanol?

A. Ethanal
B. Dimethyl ether
C. Methanol
D. Propanol
E. Ethane

Section 5 — Physics & Mathematics (Q48–60)

48. A car decelerates uniformly from 30 m/s to rest over 75 m. What is the magnitude of its deceleration?

A. 0.4 m/s^2
B. 6 m/s^2
C. 4 m/s^2
D. 12 m/s^2
E. 2 m/s^2

49. A 10 kg object is pushed up a frictionless incline to a height of 2 m. If 250 J of work is done, how much kinetic energy does it have at the top? ($g = 10\ m/s^2$)

A. 0 J
B. 250 J
C. 50 J
D. 450 J
E. 200 J

50. A 1.5 V cell is connected to a 3 Ω resistor. How much energy is transferred in 20 s?

A. 15 J
B. 1.5 J
C. 0.15 J
D. 30 J
E. 90 J

51. A pendulum has a period of 2 s. If its length is quadrupled, what is the new period?

A. 16 s
B. 8 s
C. 4 s
D. 1 s
E. 2 s

52. An object of density 800 kg/m^3 is placed in water (1000 kg/m^3). What fraction of its volume is submerged when it floats?

A. 0.2
B. 1.25
C. 1.0
D. 0.5
E. 0.8

53. A 3 kg object is in free fall. What is its weight? ($g = 10\ m/s^2$)

A. 30 N
B. 0 N
C. 10 N
D. 3 N
E. 300 N

54. What is the frequency of a wave whose period is 0.02 s?

A. 2 Hz
B. 50 Hz
C. 500 Hz
D. 0.02 Hz
E. 20 Hz

55. Simplify: log(50) + log(2) (base 10).

A. 100
B. 2
C. log 52
D. 1
E. 25

56. In triangle ABC, angle A = 90°, AB = 5 and BC = 13. What is AC?

A. 10
B. 14
C. 8
D. √194
E. 12

57. A die is rolled twice. What is the probability that at least one 6 is obtained?

A. 11/36
B. 1/3
C. 1/36
D. 25/36
E. 1/6

58. Solve the system: x + y = 7 and x − y = 1.

A. x = 6, y = 1
B. x = 3, y = 4
C. x = 4, y = 3
D. x = 2, y = 5
E. x = 5, y = 2

59. The geometric sequence 2, 6, 18, ... has what 6th term?

A. 486
B. 243
C. 729
D. 1458
E. 162

60. A wire carrying a current of 3 A lies perpendicular to a uniform magnetic field of 0.2 T. What force acts on 0.5 m of the wire?

A. 0.3 N
B. 0.03 N
C. 30 N
D. 0.6 N
E. 3 N

정답 및 해설

번호	정답	해설
1	A	질병·식단·약물이 미생물을 바꿀 수 있음 → 역인과 가능성.
2	A	원인(미생물)을 먼저 주고 결과(증상)를 관찰 → 인과 방향 확인.
3	E	리타 레비-몬탈치니(NGF).
4	C	1948년.
5	E	"일부 항생제는 바이러스에 효과" 가 거짓 → 그 부정 "어떤 항생제도 효과 없음".
6	A	1.2 × 1.1 = 1.32 → 32%.
7	B	1인 1시간 3명 → 5×6×3 = 90.
8	D	피보나치: 8+13 = 21.
9	E	400 m를 더 달려야 함. 속도 5 − 4 = 1 m/s → 400 s.
10	C	미세소관.
11	A	유전 암호의 축중성.
12	D	예정된 세포사.
13	E	아세틸-CoA + 옥살로아세트산 → 시트르산.
14	A	딸은 아버지 X(정상) + 어머니 X 중 1/2 확률로 보인자 X.
15	D	물의 광분해.
16	B	소뇌.
17	D	글리세롤 + 지방산 2 + 인산기.
18	E	랑비에 결절 사이 도약.
19	D	혈소판.
20	B	$A^O \times B^O \rightarrow OO$ 1/4.
21	C	하수체 문맥계로 방출호르몬.
22	C	G–C 4×3 = 12, A–T 6×2 = 12 → 24.
23	C	중쇄 2 + 경쇄 2.
24	E	진한 소변 생성.
25	E	제1감수분열 전기.
26	D	활성부위 아미노산의 이온화 상태 변화.
27	A	터너 증후군.
28	E	식세포작용.
29	D	역전사효소.
30	E	중탄산염이 풍부한 액.
31	D	조면 ER 리보솜 → ER 내강 → 소포 → 골지 → 분비소포 → 세포막.
32	E	DNA·단백질 서열 유사성.
33	B	6 + 12 + 6 = 24.
34	B	N + 4(+1) = +1 → −3.
35	B	$[H^+]$ $10^{-4} \rightarrow 10^{-6}$ → pH 6.

번호	정답	해설
36	B	네온.
37	D	BF_3(비공유전자쌍 없음).
38	B	$CH_4 + 2O_2 \rightarrow 2\ \text{mol} \times 22.4 = 44.8\ \text{L}$.
39	C	에탄올.
40	C	0.05 mol / 0.5 L = 0.1 mol/L.
41	C	O–H를 가진 메탄올.
42	D	아세트산 이온의 가수분해 → OH^-.
43	D	$\text{rate} = k[A][B]^2$ → 3차.
44	A	$–CONH_2$.
45	D	방위 양자수 l.
46	C	$Na^+ + e^- \rightarrow Na$.
47	B	C_2H_6O: 다이메틸 에터.
48	B	$v^2 = 2as \rightarrow 900 = 150a \rightarrow a = 6$.
49	C	$250 - mgh(200) = 50\ \text{J}$.
50	A	$P = V^2/R = 0.75\ \text{W} \times 20 = 15\ \text{J}$.
51	C	$T \propto \sqrt{L}$ → 2배.
52	E	800/1000 = 0.8.
53	A	무게 = mg = 30 N(자유낙하에서도 중력은 작용).
54	B	1/0.02 = 50 Hz.
55	B	log 100 = 2.
56	E	5-12-13.
57	A	$1 - (5/6)^2 = 11/36$.
58	C	x = 4, y = 3.
59	A	$2 \times 3^5 = 486$.
60	A	$F = BIL = 0.2 \times 3 \times 0.5 = 0.3\ \text{N}$.

IMAT 모의고사 7회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Telemedicine expanded rapidly when clinics closed their doors during the pandemic, and many health systems have kept remote consultations since. Supporters argue that video visits save patients travel time and reduce missed appointments, particularly in rural areas. Opponents warn that physical examination is impossible on a screen and that patients without reliable internet or digital skills — often older or poorer — may be left behind. A sensible policy, some suggest, would offer remote care as an option rather than a replacement.

1. Which of the following best summarises the opponents' argument?

A. Telemedicine was invented during the pandemic.
B. Missed appointments increase with telemedicine.
C. Telemedicine is more expensive than in-person care.
D. Remote care has limits and may worsen inequalities in access.
E. Rural patients prefer in-person visits.

2. The phrase "an option rather than a replacement" implies that:

A. in-person care should remain available alongside remote care.
B. all consultations should be remote.
C. internet access should be mandatory.
D. remote care should be abolished.
E. only rural patients should use remote care.

3. Which Italian city hosts the oldest university in continuous operation in the Western world, founded in 1088?

A. Rome
B. Naples
C. Bologna
D. Pisa
E. Padua

4. The Paris Agreement (2015) is an international treaty concerned mainly with:

A. nuclear disarmament
B. climate change
C. refugee rights
D. international trade tariffs
E. public health emergencies

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "Doctors who work in emergency departments are trained in resuscitation. Dr Rossi is trained in resuscitation." What can be concluded?

A. Dr Rossi is not a doctor.
B. All doctors trained in resuscitation work in emergency departments.
C. Dr Rossi works in an emergency department.
D. Dr Rossi might or might not work in an emergency department.
E. Dr Rossi does not work in an emergency department.

6. A laboratory sample contains 2400 bacteria. The population doubles every 30 minutes. How many bacteria are there after 2 hours?

A. 38 400
B. 9600
C. 19 200
D. 4800
E. 76 800

7. Six seats are arranged in a row. A sits at one end. B sits next to A. C sits at the other end. D does not sit next to C. E and F sit next to each other. Which seat (1–6, with A in seat 1) does D occupy?

A. 2
B. 5
C. 4
D. Cannot be determined
E. 3

8. What is the next term in the sequence: 1, 4, 9, 16, 25, __?

A. 49
B. 36
C. 35
D. 32
E. 30

9. A recipe for 4 people uses 300 g of rice. How much rice is needed for 10 people?

A. 700 g
B. 750 g
C. 650 g
D. 1200 g
E. 600 g

Section 3 — Biology (Q10–32)

10. Which of the following is the function of the nucleolus?

A. Protein degradation
B. Synthesis of lipids
C. Assembly of ribosomal subunits and rRNA synthesis
D. ATP production
E. DNA replication only

11. Which of the following is required for transcription in eukaryotes?
A. RNA polymerase binding to a promoter
B. Okazaki fragments
C. tRNA carrying amino acids
D. Ribosomes binding to DNA
E. DNA polymerase

12. Cancer cells commonly show:
A. absence of a nucleus.
B. uncontrolled division due to mutations in proto-oncogenes and tumour suppressor genes.
C. contact inhibition.
D. inability to replicate DNA.
E. limited division.

13. In oxidative phosphorylation, ATP synthase is driven by:
A. direct oxidation of oxygen.
B. glucose diffusion.
C. a gradient of Na^+ ions.
D. ATP hydrolysis.
E. a proton (H^+) gradient across the inner mitochondrial membrane.

14. In a family, a disease appears in every generation, affected fathers pass it to all daughters but no sons. The most likely inheritance is:
A. autosomal dominant.
B. X-linked dominant.
C. X-linked recessive.
D. autosomal recessive.
E. mitochondrial.

15. Which of the following organisms are primary producers in most ecosystems?
A. Herbivores
B. Fungi
C. Decomposers
D. Photosynthetic organisms
E. Carnivores

16. Surfactant in the alveoli functions to:
A. increase surface tension.
B. kill bacteria.
C. produce mucus.
D. reduce surface tension and prevent alveolar collapse.
E. transport oxygen.

17. Which of the following is a disaccharide composed of glucose and galactose?
A. Fructose
B. Cellulose
C. Lactose
D. Sucrose
E. Maltose

18. The refractory period of a neuron ensures that:
A. the axon becomes myelinated.
B. the neuron fires continuously.
C. K^+ ions remain inside.
D. action potentials travel in one direction.
E. neurotransmitters are reabsorbed.

19. Which of the following vessels has the thinnest walls, allowing exchange of substances?
A. Arterioles
B. Aorta
C. Arteries
D. Veins
E. Capillaries

20. In cats, the gene for orange coat (X^O) and black coat (X^B) are X-linked and codominant. A tortoiseshell (orange-and-black) cat is almost always:
A. female.
B. either sex equally.
C. male.
D. sterile.
E. homozygous.

21. Negative feedback in hormone regulation means:
A. the hormone stimulates its own further release.
B. the effect of the hormone reduces further secretion of that hormone.
C. hormone levels rise without limit.
D. the hormone is secreted only at night.
E. hormones are always inhibited.

22. If one strand of DNA has the sequence 5'-ATGCTA-3', the complementary strand is:
A. 3'-TACGAT-5'
B. 5'-ATGCTA-3'
C. 3'-ATGCTA-5'
D. 5'-TACGAT-3'
E. 3'-UACGAU-5'

23. Which cells present antigens to helper T cells using MHC class II molecules?
A. All nucleated cells only
B. Dendritic cells, macrophages and B cells
C. Erythrocytes
D. Platelets
E. Neurons

24. The juxtaglomerular apparatus releases renin when:
A. ADH levels increase.
B. blood glucose rises.
C. blood pressure falls.
D. urine is concentrated.
E. blood pressure rises.

25. Which of the following is a source of genetic variation in sexually reproducing organisms?
A. Vegetative propagation
B. Cloning
C. Binary fission
D. Mitosis
E. Independent assortment and crossing over

26. Which bond is responsible for the tertiary structure linking cysteine residues?
A. Hydrogen bond
B. Glycosidic bond
C. Disulfide bond
D. Peptide bond
E. Phosphodiester bond

27. A karyotype shows 47,XXY. This condition is called:
A. Klinefelter syndrome
B. Fragile X syndrome
C. Edwards syndrome
D. Turner syndrome
E. Down syndrome

28. Which of the following crosses the plasma membrane most easily by simple diffusion?
A. Proteins
B. O_2
C. Na^+
D. Glucose
E. Cl^-

29. Prions are infectious agents made of:
A. lipids
B. carbohydrates
C. misfolded proteins
D. DNA only
E. RNA only

30. Where is most water absorbed in the digestive tract?
A. Mouth
B. Stomach
C. Small intestine
D. Oesophagus
E. Rectum only

31. Which of the following is true of prokaryotic gene expression?
A. Ribosomes are 80S.
B. Introns are always present.
C. mRNA must be exported from a nucleus.
D. A 5' cap is added to mRNA.
E. Transcription and translation can occur simultaneously.

32. A species of moth darkened after soot polluted tree bark, then lightened after air quality improved. This illustrates:

A. genetic drift.
B. Lamarckism.
C. spontaneous generation.
D. directional natural selection.
E. artificial selection.

Section 4 — Chemistry (Q33–47)

33. What is the mass of 3×10^{23} molecules of water? (H_2O, M = 18 g/mol, N_A = 6×10^{23})

A. 6 g
B. 9 g
C. 18 g
D. 3 g
E. 36 g

34. In which of the following is the oxidation number of carbon +4?

A. CH_3OH
B. CH_4
C. CO_2
D. C_2H_6
E. CO

35. At 25 °C, a solution has pH 3. What is $[OH^-]$?

A. 10^{-14} mol/L
B. 10^{-3} mol/L
C. 10^{3} mol/L
D. 10^{-11} mol/L
E. 10^{-7} mol/L

36. Which of the following atoms has the smallest atomic radius?

A. C
B. Be
C. B
D. Li
E. F

37. Which of the following has a linear shape?

A. CH_4
B. SO_2
C. H_2O
D. $BeCl_2$
E. NH_3

38. How many grams of CO_2 are produced when 12 g of carbon burns completely in oxygen? (C = 12, O = 16)

A. 88 g
B. 12 g
C. 44 g
D. 22 g
E. 32 g

39. Which reagent distinguishes an aldehyde from a ketone by forming a silver mirror?

A. Tollens' reagent
B. Limewater
C. Universal indicator
D. Sodium chloride
E. Bromine water

40. What is the mass percentage of a solution prepared by dissolving 20 g of salt in 80 g of water?

A. 16%
B. 20%
C. 40%
D. 80%
E. 25%

41. Which statement about metallic bonding is correct?

A. Metallic bonds form between non-metals.
B. Positive ions are held in a sea of delocalised electrons.
C. Metals do not conduct electricity.
D. Electrons are fixed between two atoms.
E. Metals are always liquids.

42. Which pair is a conjugate acid–base pair?

A. NH_4^+ / NH_3
B. H_2O / H_2
C. HCl / NaOH
D. H_2SO_4 / SO_4^{2-}
E. HNO_3 / NO_2

43. Le Chatelier's principle predicts that adding more reactant to a system at equilibrium will:

A. stop the reaction.
B. shift the equilibrium towards the reactants.
C. change the equilibrium constant.
D. have no effect.
E. shift the equilibrium towards the products.

44. What is the general formula of alkanes?

A. C_nH_n
B. C_nH_{2n}
C. C_nH_{2n+2}
D. C_nH_{2n-2}
E. $C_nH_{2n+1}OH$

45. How many unpaired electrons does a ground-state nitrogen atom have?
A. 1
B. 0
C. 2
D. 5
E. 3

46. Which of the following is a reducing agent in the reaction $CuO + H_2 \rightarrow Cu + H_2O$?
A. H_2
B. H_2O
C. Cu
D. CuO
E. O^{2-}

47. Polymerisation of ethene produces:
A. polystyrene
B. nylon
C. PVC
D. polyester
E. polyethene (polyethylene)

Section 5 — Physics & Mathematics (Q48–60)

48. A body starts from rest with acceleration 2 m/s². What distance does it cover in the 3rd second only (between t = 2 s and t = 3 s)?
A. 2 m
B. 9 m
C. 4 m
D. 5 m
E. 3 m

49. A 2 kg object falls from a height of 5 m. What is its speed just before hitting the ground? (g = 10 m/s², no air resistance)
A. 5 m/s
B. 50 m/s
C. 7 m/s
D. 100 m/s
E. 10 m/s

50. What is the resistance of a lamp rated 240 V, 60 W?
A. 240 Ω
B. 14 400 Ω
C. 4 Ω
D. 0.25 Ω
E. 960 Ω

51. An object is placed 30 cm in front of a converging lens of focal length 10 cm. Where is the image formed?

A. 15 cm from the lens on the opposite side
B. No image is formed
C. 7.5 cm from the lens, same side as object
D. 20 cm from the lens on the opposite side
E. 30 cm from the lens on the opposite side

52. A gas at 2 atm occupies 3 L at 300 K. What volume does it occupy at 1 atm and 600 K?

A. 9 L
B. 6 L
C. 12 L
D. 1.5 L
E. 3 L

53. Two forces of 3 N and 4 N act on an object at right angles. What is the magnitude of the resultant force?

A. 5 N
B. 3.5 N
C. 7 N
D. 1 N
E. 12 N

54. An electric kettle of power 2000 W heats water for 3 minutes. How much energy is supplied?

A. 36 000 J
B. 360 000 J
C. 6 000 000 J
D. 6000 J
E. 120 000 J

55. Solve: $4^x = 8$.

A. 2/3
B. 1
C. 3
D. 3/2
E. 2

56. What is the value of $\tan(45°) + \sin(90°)$?

A. 2
B. $1 + \sqrt{2}/2$
C. 0
D. $\sqrt{2}$
E. 1

57. In a class, 40% of students are boys. 50% of boys and 25% of girls wear glasses. What percentage of the class wears glasses?

A. 75%
B. 30%
C. 40%
D. 35%
E. 37.5%

58. What is the area of a triangle with vertices (0, 0), (6, 0) and (0, 4)?

A. 24
B. 10
C. 20
D. 48
E. 12

59. What is the derivative of $f(x) = 3x^2 + 2x$?

A. $6x^2 + 2$
B. $3x + 2$
C. $x^3 + x^2$
D. $6x + 2$
E. $6x$

60. An X-ray photon has energy $E = hf$. If the frequency is tripled, the energy:

A. is multiplied by $\sqrt{3}$.
B. stays the same.
C. is tripled.
D. is multiplied by 9.
E. is reduced to one third.

정답 및 해설

번호	정답	해설
1	D	신체 진찰 불가 + 디지털 소외층 → 한계와 불평등 심화.
2	A	대면 진료도 계속 유지되어야 함.
3	C	볼로냐 대학(1088).
4	B	기후변화.
5	D	응급실 의사 ⊂ 소생술 훈련자 → 역은 성립 안 함.
6	A	2시간 = 4번 배가 → 2400 × 16 = 38 400.
7	E	A1, B2, C6. D는 5 불가. E·F 인접: 남은 3·4·5 중 D 위치 3이면 E·F는 4·5 (가능); D=4면 E·F 3·5 불인접 → 불가. 따라서 D=3.
8	B	$6^2 = 36$.
9	B	75 g/인 × 10 = 750 g.
10	C	rRNA 합성, 리보솜 소단위 조립.
11	A	RNA 중합효소가 프로모터에 결합.
12	B	원종양유전자·종양억제유전자 돌연변이 → 무제한 분열.
13	E	내막 사이 양성자 기울기(화학삼투).
14	B	X연관 우성: 이환 아버지 → 딸 전부, 아들 없음.
15	D	광합성 생물.
16	D	표면장력↓, 폐포 허탈 방지.
17	C	젖당 = 포도당 + 갈락토스.
18	D	불응기 → 활동전위 한 방향 전도.
19	E	모세혈관.
20	A	X 두 개 필요 → 암컷(XXY 수컷 예외).
21	B	호르몬 효과가 분비 억제.
22	A	상보·역평행: 3'-TACGAT-5'.
23	B	전문 항원제시세포(수지상세포, 대식세포, B세포).
24	C	혈압 저하 시 레닌 분비.
25	E	독립 분리 + 교차.
26	C	이황화 결합.
27	A	클라인펠터 증후군.
28	B	작은 비극성 분자 O_2.
29	C	잘못 접힌 단백질.
30	C	소장.
31	E	핵막 없음 → 전사·번역 동시 진행.
32	D	방향성 선택(공업 암화).
33	B	0.5 mol × 18 = 9 g.
34	C	CO_2: +4.
35	D	pOH 11.

번호	정답	해설
36	E	같은 주기 오른쪽일수록 작음 → F.
37	D	$BeCl_2$ 직선형.
38	C	1 mol C → 1 mol CO_2 = 44 g.
39	A	톨렌스 시약(은거울 반응).
40	B	20/(20+80) = 20%.
41	B	양이온 + 비편재 전자 바다.
42	A	NH_4^+/NH_3는 H^+ 하나 차이.
43	E	반응물 추가 → 정반응 쪽 이동. K는 불변.
44	C	C_nH_{2n+2}.
45	E	$2p^3$: 3개.
46	A	H_2가 산화됨 → 환원제.
47	E	폴리에틸렌.
48	D	s(3) − s(2) = 9 − 4 = 5 m.
49	E	v = √(2gh) = √100 = 10.
50	E	R = V^2/P = 57 600/60 = 960 Ω.
51	A	1/v = 1/10 − 1/30 = 2/30 → v = 15 cm (실상).
52	C	V_2 = 3 × (2/1) × (600/300) = 12 L.
53	A	√(9+16) = 5.
54	B	2000 × 180 = 360 000 J.
55	D	$2^{2x} = 2^3$ → x = 3/2.
56	A	1 + 1 = 2.
57	D	0.4×0.5 + 0.6×0.25 = 0.20 + 0.15 = 35%.
58	E	½ × 6 × 4 = 12.
59	D	6x + 2.
60	C	E ∝ f.

IMAT 모의고사 8회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Artificial-intelligence systems trained on thousands of retinal photographs can now detect diabetic eye disease with accuracy comparable to specialists. Advocates see this as a way to screen millions of patients in regions with few ophthalmologists. However, an algorithm that performs well on images from one hospital's cameras may perform worse on images taken with different equipment or from populations under-represented in the training data. Independent testing in real clinics, rather than reliance on developers' published figures, is therefore essential before such tools are widely adopted.

1. What is the main concern raised in the passage?

A. Retinal photographs are expensive.
B. Specialists oppose new technology.
C. AI systems are slower than ophthalmologists.
D. Diabetic eye disease is rare.
E. AI performance may not generalise beyond the conditions in which it was trained.

2. Which of the following would best address the concern described?

A. Evaluating the algorithm in diverse clinics with different equipment and patient populations.
B. Training the algorithm on images from a single hospital.
C. Publishing more of the developers' own accuracy figures.
D. Replacing all ophthalmologists.
E. Reducing the number of images used for training.

3. Which Nobel Prize-winning discovery of 1901 made it possible for physicians to see inside the body without surgery?

A. X-rays (Röntgen)
B. Insulin
C. The electron
D. Radioactivity
E. Ultrasound

4. Which of these is the official currency used in Italy?

A. Mark
B. Euro
C. Peseta
D. Lira
E. Franc

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "If the experiment is valid, then the results are reproducible. If the results are reproducible, then the paper will be accepted." Which follows?

A. The experiment is valid.
B. If the experiment is valid, the paper will be accepted.
C. If the paper is accepted, the experiment is valid.
D. If the results are not reproducible, the paper is accepted.
E. If the paper is rejected, the experiment may still be valid.

6. An investment of €2000 earns 5% simple interest per year. What is its total value after 3 years?

A. €3000
B. €2100
C. €2500
D. €2315.25
E. €2300

7. Three boxes are labelled "Apples", "Oranges" and "Mixed", but every label is wrong. You may take one fruit from one box. From which box should you take it to identify all boxes correctly?

A. "Apples"
B. "Oranges"
C. "Mixed"
D. It is impossible
E. Any box

8. What number comes next? 2, 3, 5, 7, 11, 13, __

A. 15
B. 17
C. 21
D. 19
E. 16

9. A map uses a scale of 1 : 50 000. Two towns are 6 cm apart on the map. What is the real distance?

A. 8.3 km
B. 30 km
C. 300 km
D. 3 km
E. 0.3 km

Section 3 — Biology (Q10–32)

10. Which of the following structures is found in both prokaryotic and eukaryotic cells?
A. Mitochondria
B. Ribosomes
C. Endoplasmic reticulum
D. Golgi apparatus
E. Nucleus

11. Semi-conservative replication means that:
A. both strands are newly synthesised.
B. both strands of the parent molecule remain together.
C. DNA is copied into RNA.
D. replication occurs only on one strand.
E. each new DNA molecule contains one original strand and one new strand.

12. Which phase of the cell cycle is the longest in most actively dividing human cells?
A. Metaphase
B. Interphase
C. Telophase
D. Prophase
E. Anaphase

13. Which of the following is the correct summary equation for aerobic respiration?
A. $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
B. $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy
C. $C_6H_{12}O_6 \rightarrow$ 2 lactate
D. $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
E. $C_6H_{12}O_6 + 6H_2O \rightarrow 6CO_2 + 12H_2$

14. Two heterozygous parents for cystic fibrosis (autosomal recessive) have two children. What is the probability that both children are affected?
A. 9/16
B. 1/4
C. 1/8
D. 1/2
E. 1/16

15. Which of the following best describes the role of decomposers in an ecosystem?
A. Producing oxygen
B. Predation of herbivores
C. Consuming primary producers
D. Fixing atmospheric carbon by photosynthesis
E. Breaking down dead organic matter and recycling nutrients

16. The cells lining the trachea that sweep mucus upwards have:
A. keratin.
B. cilia.
C. flagella.
D. desmosomes only.
E. microvilli.

17. Which of the following elements is present in all proteins but not in carbohydrates?
A. Carbon
B. Phosphorus
C. Hydrogen
D. Nitrogen
E. Oxygen

18. Which part of a neuron receives signals from other neurons?
A. Axon terminal
B. Dendrites
C. Myelin sheath
D. Axon hillock only
E. Node of Ranvier

19. In the ECG, the QRS complex corresponds to:
A. ventricular depolarisation.
B. atrial repolarisation only.
C. closing of the semilunar valves.
D. atrial depolarisation.
E. ventricular repolarisation.

20. A plant heterozygous for three independently assorting genes (AaBbCc) is self-fertilised. What proportion of offspring is homozygous recessive for all three genes?
A. 1/16
B. 1/64
C. 27/64
D. 1/27
E. 1/8

21. Which hormone triggers ovulation?
A. Luteinising hormone (LH)
B. Testosterone
C. Progesterone
D. Oestrogen
E. Prolactin

22. Which of the following describes Chargaff's rule?
A. A = T and G = C
B. A = G and C = T
C. Purines = 25% always
D. A + T = G + C always
E. A = U in DNA

23. Which of the following is an example of passive immunity?
A. Clonal expansion of B cells
B. Maternal antibodies transferred through breast milk
C. Production of memory cells
D. Immunity after vaccination
E. Immunity after recovering from measles

24. Which of the following would be expected in the urine of a person with uncontrolled diabetes mellitus?

A. Proteins
B. Red blood cells
C. Glucose
D. Bilirubin only
E. Platelets

25. What is the number of chromosomes in a human secondary oocyte?

A. 23
B. 22
C. 92
D. 46
E. 69

26. The induced-fit model states that:

A. substrates change enzymes permanently.
B. enzymes are not specific.
C. the active site is rigid and fits the substrate exactly.
D. enzymes have no active site.
E. the active site changes shape slightly when the substrate binds.

27. A point mutation changing GAA (Glu) to GAG (Glu) is:

A. a chromosomal mutation.
B. a frameshift mutation.
C. a nonsense mutation.
D. a silent mutation.
E. a missense mutation.

28. Which of these is NOT a component of the plasma membrane?

A. Peptidoglycan
B. Glycolipids
C. Cholesterol
D. Proteins
E. Phospholipids

29. Bacteriophages are:

A. bacteria that infect viruses.
B. prions.
C. viruses that infect bacteria.
D. protists.
E. fungi.

30. Which organ produces most digestive enzymes that act in the small intestine?

A. Liver
B. Spleen
C. Gallbladder
D. Pancreas
E. Stomach

31. In eukaryotes, where does transcription occur?

A. Golgi apparatus
B. Cytoplasm
C. Nucleus
D. Lysosome
E. Ribosome

32. Which of the following conditions is NOT required for Hardy–Weinberg equilibrium?

A. No mutation
B. No gene flow
C. Random mating
D. Strong natural selection
E. Large population size

Section 4 — Chemistry (Q33–47)

33. What volume does 0.5 mol of an ideal gas occupy at STP (22.4 L/mol)?

A. 44.8 L
B. 2.24 L
C. 11.2 L
D. 22.4 L
E. 5.6 L

34. What is the oxidation number of oxygen in H_2O_2?

A. +2
B. −2
C. +1
D. 0
E. −1

35. Which of the following solutions has the lowest pH?

A. 0.1 mol/L HCl
B. 0.1 mol/L NH_3
C. 0.1 mol/L CH_3COOH
D. 0.1 mol/L NaOH
E. 0.1 mol/L NaCl

36. Which element is a halogen in period 3?

A. Na
B. S
C. Cl
D. Br
E. Ar

37. What is the shape of the methane molecule?

A. Trigonal pyramidal
B. Octahedral
C. Tetrahedral
D. Linear
E. Square planar

38. In the reaction $2KClO_3 \rightarrow 2KCl + 3O_2$, how many moles of O_2 are formed from 4 mol of $KClO_3$?

A. 6
B. 12
C. 3
D. 4
E. 2

39. What is the IUPAC name of $CH_3COOCH_2CH_3$?

A. Butanoic acid
B. Methyl propanoate
C. Ethyl methanoate
D. Ethyl ethanoate
E. Propyl methanoate

40. A 0.2 mol/L solution of $CaCl_2$ contains what concentration of Cl^- ions?

A. 0.1 mol/L
B. 0.8 mol/L
C. 0.2 mol/L
D. 0.6 mol/L
E. 0.4 mol/L

41. Which of the following is a polar molecule?

A. CCl_4
B. CO_2
C. N_2
D. HCl
E. CH_4

42. Which of the following is a Lewis acid?

A. H_2O
B. Cl^-
C. BF_3
D. NH_3
E. OH^-

43. For an exothermic reaction, increasing temperature will:

A. stop the reaction.
B. increase the equilibrium constant.
C. not change the equilibrium constant.
D. decrease the equilibrium constant.
E. make it endothermic.

44. Which pair of compounds are functional group isomers?

A. Ethanol and methanol
B. Propan-1-ol and propan-2-ol
C. Propanal and propanone
D. Butane and methylpropane
E. cis- and trans-but-2-ene

45. How many neutrons are in $^{23}Na^{+}$?

A. 12
B. 23
C. 10
D. 11
E. 22

46. Rusting of iron is an example of:

A. reduction of iron.
B. sublimation.
C. oxidation of iron.
D. neutralisation.
E. precipitation.

47. Which of the following is a saturated hydrocarbon?

A. Propane
B. Ethyne
C. Benzene
D. Propene
E. Ethene

Section 5 — Physics & Mathematics (Q48–60)

48. A stone is thrown upward at 15 m/s. How long does it take to return to the thrower's hand? (g = 10 m/s²)

A. 2 s
B. 1.5 s
C. 4.5 s
D. 3 s
E. 6 s

49. A motor lifts a 50 kg load at constant speed of 2 m/s. What is the motor's power output? (g = 10 m/s²)

A. 500 W
B. 250 W
C. 2000 W
D. 1000 W
E. 100 W

50. In a circuit, a 6 V battery drives current through a 2 Ω and a 4 Ω resistor in series. What is the voltage across the 4 Ω resistor?

A. 3 V
B. 6 V
C. 2 V
D. 1 V
E. 4 V

51. Which type of electromagnetic radiation has the shortest wavelength?

A. Microwaves
B. Radio waves
C. Ultraviolet
D. Visible light
E. Gamma rays

52. A diver goes from the surface to 20 m depth in water. By how much does the pressure increase? ($\rho = 1000$ kg/m³, $g = 10$ m/s²)

A. 2×10^5 Pa
B. 2×10^6 Pa
C. 2×10^3 Pa
D. 10^5 Pa
E. 2×10^4 Pa

53. A net force of 10 N acts on a 2 kg object for 3 s, starting from rest. What is the object's final speed?

A. 5 m/s
B. 10 m/s
C. 15 m/s
D. 6.7 m/s
E. 30 m/s

54. How much energy is required to melt 0.5 kg of ice at 0 °C? (Latent heat of fusion = 3.3×10^5 J/kg)

A. 1.65×10^4 J
B. 3.3×10^5 J
C. 6.6×10^5 J
D. 6.6×10^4 J
E. 1.65×10^5 J

55. What is the value of x if $\ln(x) = 0$?

A. 10
B. 1
C. e
D. -1
E. 0

56. Convert $3\pi/4$ radians to degrees.

A. 270°
B. 90°
C. 45°
D. 135°
E. 120°

57. A test for a disease is positive in 90% of patients with the disease. In a sample of 200 patients with the disease, how many would be expected to test negative?

A. 90
B. 18
C. 180
D. 20
E. 10

58. What is the volume of a sphere of radius 3 cm?

A. 12π cm³
B. 36π cm³
C. 27π cm³
D. 9π cm³
E. 108π cm³

59. Factorise: $x^2 + x - 12$.

A. $(x + 3)(x - 4)$
B. $(x - 6)(x + 2)$
C. $(x - 3)(x + 4)$
D. $(x + 6)(x - 2)$
E. $(x + 12)(x - 1)$

60. An alpha particle consists of:

A. two electrons.
B. one proton.
C. one neutron.
D. one electron.
E. two protons and two neutrons.

정답 및 해설

번호	정답	해설
1	E	학습 환경 밖에서의 일반화 문제.
2	A	다양한 임상 환경에서 독립 검증.
3	A	뢴트겐 X선(1901 첫 노벨 물리학상).
4	B	유로.
5	B	삼단 연쇄(가언 삼단논법).
6	E	2000 × 0.05 × 3 = 300 → 2300.
7	C	"Mixed" 라벨 상자는 반드시 단일 과일 → 하나 꺼내면 그 상자 확정, 나머지도 연쇄적으로 결정.
8	B	소수열 → 17.
9	D	6 × 50 000 = 300 000 cm = 3 km.
10	B	리보솜.
11	E	반보존적 복제.
12	B	간기.
13	B	호흡 반응식.
14	E	$(1/4)^2 = 1/16$.
15	E	사체 분해, 영양소 순환.
16	B	섬모.
17	D	질소.
18	B	가지돌기.
19	A	심실 탈분극.
20	B	$(1/4)^3 = 1/64$.
21	A	LH 급증.
22	A	A = T, G = C.
23	B	모유 항체(수동면역).
24	C	포도당뇨.
25	A	제1감수분열 후 n = 23.
26	E	기질 결합 시 활성부위 형태가 약간 변함.
27	D	같은 Glu → 침묵 돌연변이.
28	A	펩티도글리칸은 세균 세포벽.
29	C	세균을 감염하는 바이러스.
30	D	췌장.
31	C	핵.
32	D	자연선택이 없어야 함.
33	C	0.5 × 22.4 = 11.2 L.
34	E	과산화물 −1.
35	A	강산 0.1 mol/L HCl → pH 1.

번호	정답	해설
36	C	염소.
37	C	정사면체.
38	A	2 : 3 → 4 : 6.
39	D	아세트산에틸(에틸 에타노에이트).
40	E	Cl^- 2개 × 0.2 = 0.4.
41	D	HCl.
42	C	BF_3는 전자쌍 수용체.
43	D	발열 반응: T↑ → K↓.
44	C	알데하이드 vs 케톤(C_3H_6O).
45	A	23 − 11 = 12.
46	C	Fe → Fe^{3+} 산화.
47	A	프로페인.
48	D	올라가는 시간 1.5 s × 2 = 3 s.
49	D	P = mgv = 50×10×2 = 1000 W.
50	E	I = 1 A → 4 Ω에 4 V.
51	E	감마선.
52	A	ρgh = 1000×10×20 = 2×10^5 Pa.
53	C	a = 5 → v = 15 m/s.
54	E	0.5 × 3.3×10^5 = 1.65×10^5 J.
55	B	e^0 = 1.
56	D	135°.
57	D	10% × 200 = 20.
58	B	4/3 π × 27 = 36π.
59	C	(x−3)(x+4).
60	E	양성자 2 + 중성자 2(헬륨 핵).

IMAT 모의고사 9회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Many countries have introduced taxes on sugar-sweetened beverages. Early evaluations show that purchases of taxed drinks fell, and some manufacturers reduced the sugar content of their products to avoid the tax. Whether these changes translate into lower rates of obesity is harder to establish, because obesity develops slowly and is influenced by many factors. Critics also argue that such taxes fall more heavily on low-income households, although supporters reply that those same households bear a disproportionate share of diet-related disease.

1. According to the passage, which effect of sugar taxes is most clearly established?

A. Elimination of diet-related disease
B. Higher incomes for poorer households
C. A large fall in obesity rates
D. An increase in sugar content
E. A reduction in purchases of taxed drinks

2. How do supporters respond to the criticism that the taxes affect low-income households more?

A. They argue that poorer households also suffer more from diet-related illness and thus may benefit more.
B. They deny that poorer households buy sugary drinks.
C. They claim the tax has no effect on prices.
D. They propose abolishing the tax.
E. They say obesity is not influenced by diet.

3. Who proposed the theory of evolution by natural selection, published in *On the Origin of Species* (1859)?

A. Gregor Mendel
B. Jean-Baptiste Lamarck
C. Carl Linnaeus
D. Charles Darwin
E. Thomas Malthus

4. Which body of the European Union is directly elected by citizens of member states?

A. The Court of Justice
B. The European Parliament
C. The European Commission
D. The European Council
E. The European Central Bank

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "Either the sample was contaminated, or the machine is faulty. The machine is not faulty." Therefore:

A. Both are true.
B. Nothing follows.
C. Neither is true.
D. The sample was contaminated.
E. The sample was not contaminated.

6. A tablet contains 250 mg of active ingredient. A patient needs 1.5 g per day, divided equally into 3 doses. How many tablets per dose?

A. 2
B. 6
C. 3
D. 4
E. 1

7. In a tournament, each of 6 teams plays every other team exactly once. How many matches are played?

A. 15
B. 12
C. 36
D. 6
E. 30

8. If all Blips are Zogs and some Zogs are Fems, which statement must be true?

A. All Fems are Zogs.
B. No Blips are Fems.
C. All Blips are Fems.
D. Some Zogs are Blips (assuming Blips exist).
E. Some Blips are Fems.

9. The ratio of boys to girls in a class is 3 : 4. If there are 12 boys, how many students are there in total?

A. 16
B. 24
C. 21
D. 36
E. 28

Section 3 — Biology (Q10–32)

10. Which organelle is responsible for the detoxification of drugs in liver cells?

A. Smooth ER
B. Rough ER
C. Centrosome
D. Ribosome
E. Nucleolus

11. Which enzyme unwinds the DNA double helix at the replication fork?

A. Primase
B. DNA polymerase
C. Telomerase
D. Ligase
E. Helicase

12. During which phase does the nuclear envelope reform around the chromosomes?

A. S phase
B. Metaphase
C. Telophase
D. Prophase
E. Anaphase

13. Which of the following yields energy in the form of ATP during glycolysis?

A. The Calvin cycle
B. Photophosphorylation
C. β-oxidation
D. Substrate-level phosphorylation
E. Oxidative phosphorylation

14. Achondroplasia is autosomal dominant. An affected heterozygous man and an unaffected woman have a child. What is the probability that the child is affected?

A. 3/4
B. 1/2
C. 1
D. 0
E. 1/4

15. Which of the following is the correct sequence in a food chain?

A. Decomposer → producer → herbivore
B. Secondary consumer → producer → primary consumer
C. Producer → primary consumer → secondary consumer
D. Herbivore → producer → carnivore
E. Carnivore → herbivore → producer

16. Which muscle is the main muscle of inspiration?

A. Diaphragm
B. Internal intercostals
C. Abdominal muscles
D. Biceps
E. Trapezius

17. Which of the following carbohydrates forms the structural component of plant cell walls?

A. Cellulose
B. Lactose
C. Starch
D. Chitin
E. Glycogen

18. Which statement about the autonomic nervous system is correct?

A. The parasympathetic system prepares the body for "fight or flight".
B. The sympathetic system decreases heart rate.
C. The sympathetic system dilates the pupils and increases heart rate.
D. The autonomic system controls skeletal muscles voluntarily.
E. The parasympathetic system increases blood glucose.

19. Systemic blood pressure is highest in the:

A. capillaries.
B. venules.
C. veins.
D. vena cava.
E. aorta.

20. A cross produces offspring in a 9 : 3 : 4 ratio. This is an example of:

A. sex linkage.
B. codominance.
C. incomplete dominance.
D. epistasis.
E. linkage without crossing over.

21. Which hormone is produced by the corpus luteum after ovulation?

A. FSH
B. Progesterone
C. LH
D. Prolactin
E. Oxytocin

22. Which of the following is a purine?

A. Cytosine
B. Ribose
C. Uracil
D. Thymine
E. Adenine

23. Which antibody class is the first to be produced in a primary immune response?

A. IgE
B. IgD
C. IgG
D. IgA
E. IgM

24. Which of the following substances is secreted (rather than filtered) into the tubule in the kidney?

A. Glucose
B. Red blood cells
C. Albumin
D. H^+ ions
E. Amino acids

25. How many genetically different gametes can an individual with genotype AaBbCcDd produce (independent assortment)?

A. 16
B. 8
C. 32
D. 4
E. 64

26. An enzyme's Vmax is reached when:

A. a competitive inhibitor is added.
B. the temperature is 0 °C.
C. the enzyme is denatured.
D. the substrate concentration is zero.
E. all active sites are saturated with substrate.

27. Which type of mutation involves a segment of chromosome breaking off and reattaching in reversed orientation?

A. Duplication
B. Deletion
C. Inversion
D. Substitution
E. Translocation

28. Plant cells placed in a hypotonic solution become:

A. crenated.
B. plasmolysed.
C. turgid.
D. lysed.
E. flaccid.

29. Which of the following is a eukaryotic pathogen?

A. Streptococcus pyogenes
B. Plasmodium falciparum
C. Influenza virus
D. HIV
E. Escherichia coli

30. The villi of the small intestine increase:

A. peristalsis in the oesophagus.
B. acid production.
C. the surface area for absorption.
D. the time food spends in the stomach.
E. bile production.

31. How many different codons can be formed from the four RNA bases?

A. 4
B. 64
C. 20
D. 16
E. 61

32. Which of the following is an example of convergent evolution?

A. Darwin's finches
B. Homologous genes in mammals
C. Antibiotic resistance
D. The wings of bats and the wings of insects
E. The forelimbs of humans and whales

Section 4 — Chemistry (Q33–47)

33. Which sample contains the greatest number of atoms?

A. 1 mol of H_2O
B. 1 mol of CH_4
C. 1 mol of NaCl
D. 1 mol of He
E. 1 mol of O_2

34. In the reaction $Cl_2 + 2NaBr \rightarrow 2NaCl + Br_2$, which statement is correct?

A. Cl_2 is oxidised.
B. No redox occurs.
C. Br^- is reduced.
D. Cl_2 is the oxidising agent.
E. Na^+ is oxidised.

35. What is the pH of a 0.005 mol/L solution of H_2SO_4 (assume complete dissociation of both protons)?

A. 3
B. 2.3
C. 2
D. 1
E. 5

36. Which of the following groups contains the alkaline earth metals?

A. Group 17
B. Group 2
C. Group 13
D. Group 18
E. Group 1

37. How many lone pairs are on the central atom of H_2O?

A. 1
B. 0
C. 3
D. 4
E. 2

38. 5.6 g of iron (Fe = 56) reacts completely with sulfur: Fe + S → FeS. What mass of FeS (M = 88 g/mol) is formed?

A. 5.6 g
B. 8.8 g
C. 3.2 g
D. 88 g
E. 17.6 g

39. Which compound will react with sodium hydrogencarbonate to release CO_2?

A. Ethanal
B. Propanone
C. Ethanol
D. Ethane
E. Ethanoic acid

40. How many moles of NaOH are needed to neutralise 50 mL of 0.2 mol/L H_2SO_4?

A. 0.04
B. 0.02
C. 0.1
D. 0.005
E. 0.01

41. Which of the following has the lowest boiling point?

A. CH_4
B. C_2H_5OH
C. HF
D. H_2O
E. NH_3

42. Which of the following solutions is neutral at 25 °C?

A. $[H^+] = 10^{-5}$ mol/L
B. pOH = 4
C. pH = 9
D. $[OH^-] = 10^{-3}$ mol/L
E. $[H^+] = 10^{-7}$ mol/L

43. The half-life of a first-order reaction:

A. is independent of the initial concentration.
B. depends on the initial concentration.
C. is zero.
D. doubles each time.
E. is always 1 s.

44. What type of isomerism is shown by the two forms of lactic acid (2-hydroxypropanoic acid)?

A. No isomerism
B. Optical isomerism
C. Chain isomerism
D. Structural isomerism
E. Geometric (cis–trans) isomerism

45. Which element has the highest electronegativity?
A. C
B. O
C. F
D. Cl
E. N

46. How many electrons are transferred when one Fe^{3+} ion is reduced to Fe?
A. 6
B. 1
C. 3
D. 2
E. 4

47. Which reaction is characteristic of alkenes but not alkanes?
A. Cracking
B. Dissolving in water
C. Electrophilic addition of Br_2 (decolourising bromine water)
D. Combustion
E. Free-radical substitution

Section 5 — Physics & Mathematics (Q48–60)

48. The velocity–time graph of an object is a straight line from 0 to 12 m/s over 4 s. What is the distance travelled?
A. 48 m
B. 12 m
C. 24 m
D. 6 m
E. 3 m

49. A pump does 6000 J of useful work while consuming 8000 J of energy. What is its efficiency?
A. 133%
B. 80%
C. 25%
D. 75%
E. 60%

50. Two identical 10 Ω resistors are connected in parallel to a 5 V supply. What is the total current?
A. 0.5 A
B. 2 A
C. 5 A
D. 0.25 A
E. 1 A

51. A note has a frequency of 440 Hz. What is its period?

A. 0.00227 s
B. 440 s
C. 2.27 s
D. 0.227 s
E. 0.0227 s

52. A block of volume 0.002 m^3 is fully submerged in water. What is the buoyant force? (ρ = 1000 kg/m^3, g = 10 m/s^2)

A. 2 N
B. 2000 N
C. 20 N
D. 10 N
E. 200 N

53. A 70 kg astronaut on the Moon (g = 1.6 m/s^2) has a weight of:

A. 70 N
B. 700 N
C. 43.75 N
D. 112 N
E. 1120 N

54. The temperature 27 °C is equal to:

A. 327 K
B. 300 K
C. 273 K
D. 246 K
E. 27 K

55. If log_a(8) = 3, what is a?

A. 24
B. 1/2
C. 8/3
D. 3
E. 2

56. For which value of x in [0°, 360°) is sin x = −1?

A. 270°
B. 180°
C. 360°
D. 90°
E. 0°

57. Two cards are drawn without replacement from a standard 52-card deck. What is the probability that both are aces?

A. 1/221
B. 1/169
C. 1/26
D. 4/52
E. 1/13

58. What is the gradient of the line $4x + 2y = 6$?

A. 2
B. -2
C. 3
D. $-1/2$
E. 4

59. If $x = 2$ is a root of $x^2 + kx - 10 = 0$, what is k?

A. 7
B. 2
C. 5
D. 3
E. -3

60. Which of the following is a vector quantity?

A. Temperature
B. Speed
C. Energy
D. Displacement
E. Mass

정답 및 해설

번호	정답	해설
1	E	과세 음료 구매 감소가 확인된 효과.
2	A	저소득층이 식이 관련 질환 부담도 더 크므로 더 큰 이득.
3	D	다윈.
4	B	유럽의회.
5	D	선언 삼단논법(선언지 제거).
6	A	1500/3 = 500 mg → 2정.
7	A	${}_6C_2 = 15$.
8	D	Blip ⊂ Zog → Blip이 존재하면 일부 Zog는 Blip.
9	E	12 → 3배 → 3+4 = 7 × 4 = 28.
10	A	활면소포체(시토크롬 P450).
11	E	헬리케이스.
12	C	말기.
13	D	기질 수준 인산화.
14	B	Aa × aa → 1/2.
15	C	생산자 → 1차 소비자 → 2차 소비자.
16	A	횡격막.
17	A	셀룰로스.
18	C	교감: 동공 확대, 심박↑.
19	E	대동맥.
20	D	열성 상위(9:3:4).
21	B	프로게스테론.
22	E	아데닌(퓨린: A, G).
23	E	IgM.
24	D	H^+ 분비.
25	A	$2^4 = 16$.
26	E	활성부위 포화.
27	C	역위.
28	C	팽압 상태(turgid).
29	B	말라리아원충(원생생물).
30	C	흡수 표면적 증가.
31	B	$4^3 = 64$.
32	D	박쥐 날개 vs 곤충 날개(상사기관).
33	B	CH_4 원자 5개/분자.
34	D	Cl_2 환원 → 산화제.
35	C	$[H^+] = 0.01$ → pH 2.

번호	정답	해설
36	B	2족.
37	E	2개.
38	B	0.1 mol → 8.8 g.
39	E	카복실산 + $NaHCO_3$ → CO_2.
40	B	H_2SO_4 0.01 mol × 2 = 0.02 mol.
41	A	CH_4(수소결합 없음, 분자량 작음).
42	E	$[H^+] = 10^{-7}$.
43	A	$t_{1/2} = \ln 2/k$, 초기 농도 무관.
44	B	키랄 탄소 → 광학 이성질.
45	C	플루오린.
46	C	3개.
47	C	브로민수 탈색(친전자성 첨가).
48	C	삼각형 면적 ½×4×12 = 24 m.
49	D	6000/8000 = 75%.
50	E	5 Ω → 1 A.
51	A	1/440 ≈ 0.00227 s.
52	C	ρVg = 1000×0.002×10 = 20 N.
53	D	70 × 1.6 = 112 N.
54	B	27 + 273 = 300 K.
55	E	$2^3 = 8$.
56	A	270°.
57	A	4/52 × 3/51 = 12/2652 = 1/221.
58	B	$y = -2x + 3$.
59	D	$4 + 2k - 10 = 0 \rightarrow k = 3$.
60	D	변위.

IMAT 모의고사 10회 (고난도)

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> When a new drug is approved, its benefits are usually reported as a relative risk reduction: "cuts the risk of stroke by 50%". This figure can sound impressive even when the absolute benefit is small. If a stroke would occur in 2 of every 100 untreated patients and in 1 of every 100 treated patients, the relative reduction is 50%, but the absolute reduction is only one percentage point — meaning 100 patients must be treated for one to benefit. Communicating both numbers, some argue, would allow patients to weigh benefits against side effects more realistically.

1. Based on the passage, if a drug reduces the risk of an event from 4% to 3%, what is the "number needed to treat" for one patient to benefit?

A. 1
B. 100
C. 75
D. 25
E. 4

2. The author implies that reporting only relative risk reduction may:

A. be illegal.
B. understate the benefits of drugs.
C. be the most accurate way to describe a drug.
D. make the benefits of a drug appear larger than they are in practical terms.
E. increase side effects.

3. Which scientist formulated the laws of inheritance by studying pea plants in the 19th century?

A. Francis Crick
B. Thomas Hunt Morgan
C. Gregor Mendel
D. Hugo de Vries
E. Charles Darwin

4. The Hippocratic Oath is traditionally associated with which ancient civilisation?

A. Persian
B. Roman
C. Egyptian
D. Chinese
E. Greek

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "No drug that causes drowsiness is allowed for pilots on duty. Some antihistamines cause drowsiness." Which conclusion follows?

A. No antihistamines are allowed for pilots on duty.
B. Some drugs that cause drowsiness are allowed.
C. All drugs not allowed for pilots cause drowsiness.
D. All antihistamines are allowed for pilots on duty.
E. Some antihistamines are not allowed for pilots on duty.

6. Solution A has a concentration of 10% and solution B of 30%. In what ratio (A : B) must they be mixed to obtain a 15% solution?

A. 3 : 2
B. 1 : 3
C. 2 : 1
D. 3 : 1
E. 1 : 1

7. A clock shows 3:00. What is the smaller angle between the hour and minute hands at 3:30?

A. 120°
B. 90°
C. 75°
D. 60°
E. 105°

8. Five numbers have a mean of 10 and a median of 8. If the largest number is increased by 10, what are the new mean and median?

A. Mean 10, median 10
B. Mean 20, median 18
C. Mean 12, median 10
D. Mean 10, median 8
E. Mean 12, median 8

9. Water evaporates from a pool at a rate of 2% of the current volume per day. Approximately what fraction of the original volume remains after 2 days?

A. 0.9216
B. 0.96
C. 0.98
D. 0.9604
E. 0.94

Section 3 — Biology (Q10–32)

10. A protein is found to have a signal sequence and ends up in the lumen of the lysosome. Which tag is responsible for sorting it to lysosomes in the Golgi?

A. Ubiquitin
B. Nuclear localisation signal
C. KDEL sequence
D. Mannose-6-phosphate
E. Poly-A tail

11. Telomerase is particularly active in which of the following?

A. Mature neurons
B. Platelets
C. Red blood cells
D. Germ cells and most cancer cells
E. Muscle fibres

12. A drug blocks the formation of the mitotic spindle. In which phase will cells accumulate?

A. Metaphase (M phase)
B. G2
C. G0
D. G1
E. S

13. Cyanide inhibits cytochrome c oxidase (complex IV). What is the immediate effect?

A. The Krebs cycle speeds up.
B. Glycolysis stops completely.
C. Oxygen consumption increases.
D. Electron transport and oxidative ATP production stop.
E. The proton gradient increases.

14. A man affected by an X-linked recessive disorder marries a carrier woman. What is the probability that a daughter is affected?

A. 1
B. 1/2
C. 0
D. 3/4
E. 1/4

15. In C_4 plants, CO_2 is initially fixed into:

A. a 4-carbon compound by PEP carboxylase.
B. glucose directly.
C. oxygen.
D. a 3-carbon compound by RuBisCO.
E. ATP.

16. Which of the following best explains why carbon monoxide is dangerous?

A. It blocks the trachea.
B. It destroys red blood cells.
C. It dissolves in plasma and lowers pH.
D. It increases ventilation rate.
E. It binds haemoglobin with much higher affinity than O_2, reducing oxygen transport.

17. Which molecule is the immediate energy source for muscle contraction?

A. Fatty acids
B. ATP
C. Creatine
D. Glycogen
E. Glucose

18. A drug blocks voltage-gated K^+ channels in neurons. Which effect is expected?

A. Faster repolarisation
B. No depolarisation
C. Hyperpolarisation at rest
D. Increased myelination
E. Prolonged action potentials due to slowed repolarisation

19. Starling forces at the arterial end of a capillary favour:

A. filtration of fluid out of the capillary due to higher hydrostatic pressure.
B. movement of red blood cells into tissues.
C. no fluid movement.
D. movement of proteins out of the capillary.
E. reabsorption of fluid due to high oncotic pressure.

20. Two genes are 20 map units apart. A double heterozygote AB/ab is test-crossed. What proportion of offspring will be Ab?

A. 80%
B. 20%
C. 40%
D. 5%
E. 10%

21. Which of the following is correct about hormones derived from cholesterol?

A. They are stored in secretory granules for long periods.
B. They bind to receptors on the cell surface only.
C. They are peptides.
D. They can cross the plasma membrane and often bind intracellular receptors.
E. They are water-soluble and travel freely in plasma.

22. A DNA sample melts (denatures) at a higher temperature than another. The first sample most likely has:

A. a higher proportion of A–T base pairs.
B. a higher proportion of G–C base pairs.
C. single-stranded regions only.
D. more uracil.
E. fewer phosphate groups.

23. Rhesus disease of the newborn can occur when:
A. both parents are Rh^-.
B. the mother has blood group AB.
C. the fetus is Rh^- and the father is Rh^-.
D. an Rh^+ mother carries an Rh^- fetus.
E. an Rh^- mother, previously sensitised, carries an Rh^+ fetus.

24. A loop diuretic inhibits the Na^+–K^+–$2Cl^-$ cotransporter. Where does it act?
A. Descending limb
B. Glomerulus
C. Collecting duct
D. Thick ascending limb of the loop of Henle
E. Proximal convoluted tubule

25. A cell undergoing meiosis has 4 chromosomes in G1 (2n = 4). How many DNA molecules (chromatids) are present in each cell at metaphase II?
A. 8
B. 1
C. 4
D. 16
E. 2

26. In an enzyme-catalysed reaction, a non-competitive inhibitor:
A. has no effect on the reaction rate.
B. decreases Vmax without changing Km.
C. binds only to the active site.
D. increases Vmax.
E. increases Km without changing Vmax.

27. A mutation in the promoter region of a gene is most likely to affect:
A. the rate of transcription of the gene.
B. tRNA charging.
C. the stop codon.
D. splicing of introns.
E. the amino acid sequence of the protein.

28. Which of the following explains how glucose is absorbed into intestinal epithelial cells against its concentration gradient?
A. Secondary active transport coupled to Na^+ (SGLT1)
B. Osmosis
C. Endocytosis
D. Primary active transport using ATP directly on glucose
E. Simple diffusion

29. Penicillin is ineffective against Mycoplasma bacteria because they:
A. are viruses.
B. lack ribosomes.
C. are too large.
D. have a nucleus.
E. lack a peptidoglycan cell wall.

30. Which hormone, released from the duodenum in response to fats, stimulates gallbladder contraction?

A. Somatostatin
B. Secretin
C. Insulin
D. Gastrin
E. Cholecystokinin (CCK)

31. Alternative splicing allows:

A. tRNA to recognise several codons.
B. prokaryotes to remove introns.
C. one gene to code for multiple different proteins.
D. DNA to replicate faster.
E. ribosomes to read mRNA in reverse.

32. In a population of 1000 individuals in Hardy–Weinberg equilibrium, 90 show the recessive phenotype. How many individuals are expected to be heterozygous?

A. 490
B. 210
C. 700
D. 420
E. 90

Section 4 — Chemistry (Q33–47)

33. A hydrocarbon contains 85.7% carbon by mass and has a molar mass of 56 g/mol. What is its molecular formula? (C = 12, H = 1)

A. C_3H_6
B. C_4H_8
C. C_2H_4
D. C_5H_{12}
E. C_4H_{10}

34. Balance in acidic solution: $Cr_2O_7^{2-} + Fe^{2+} \rightarrow Cr^{3+} + Fe^{3+}$. How many H^+ ions appear on the reactant side?

A. 7
B. 8
C. 16
D. 14
E. 6

35. 25 mL of 0.1 mol/L NaOH is titrated with 0.1 mol/L HCl. After 24.9 mL of HCl has been added, the solution is:

A. strongly acidic.
B. a buffer.
C. strongly acidic with excess HCl.
D. exactly neutral.
E. slightly basic.

36. Which of the following ions has the same electron configuration as argon?
A. Mg^{2+}
B. K^{+}
C. F^{-}
D. Li^{+}
E. Na^{+}

37. What is the hybridisation of the carbon atoms in ethyne (C_2H_2)?
A. sp^2
B. sp
C. sp^3d
D. sp^3
E. sp^3d^2

38. 2.4 g of magnesium (Mg = 24) reacts with excess HCl: $Mg + 2HCl \rightarrow MgCl_2 + H_2$. What volume of H_2 is produced at STP (22.4 L/mol)?
A. 1.12 L
B. 2.24 L
C. 0.224 L
D. 4.48 L
E. 22.4 L

39. Which of the following alcohols cannot be oxidised by acidified potassium dichromate?
A. Methanol
B. Ethanol
C. 2-methylpropan-2-ol
D. Propan-2-ol
E. Propan-1-ol

40. What mass of NaOH (M = 40 g/mol) is needed to prepare 200 mL of a solution with pH 13 at 25 °C?
A. 4 g
B. 0.4 g
C. 0.8 g
D. 0.08 g
E. 8 g

41. Which of the following has the highest lattice energy?
A. KCl
B. MgO
C. KBr
D. NaCl
E. NaBr

42. The pKa of acetic acid is 4.76. A buffer contains 0.1 mol/L acetic acid and 1.0 mol/L sodium acetate. What is its pH?
A. 3.76
B. 4.76
C. 5.76
D. 6.76
E. 7.00

43. For the reaction $2NO_2(g) \rightleftharpoons N_2O_4(g)$, Kc = 4 at a given temperature. What is Kc for $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ at the same temperature?

A. 4
B. −4
C. 0.25
D. 16
E. 2

44. How many stereoisomers does a molecule with two different chiral centres (and no internal symmetry) have at most?

A. 4
B. 1
C. 8
D. 3
E. 2

45. An element X forms an oxide X_2O_3. Which group is X most likely in?

A. Group 2
B. Group 1
C. Group 13
D. Group 17
E. Group 15

46. In a concentration cell, both electrodes are copper in $CuSO_4$ solutions of 1 mol/L and 0.01 mol/L. Electrons flow in the external circuit:

A. from the 0.01 mol/L electrode to the 1 mol/L electrode.
B. from the 1 mol/L electrode to the 0.01 mol/L electrode.
C. only if a salt bridge is absent.
D. in both directions equally.
E. not at all.

47. Hydrolysis of a triglyceride with NaOH (saponification) produces:

A. esters and water.
B. glucose and fatty acids.
C. glycerol and fatty acid sodium salts (soap).
D. ethanol and acetic acid.
E. amino acids.

Section 5 — Physics & Mathematics (Q48–60)

48. A ball is thrown at 20 m/s at 30° above the horizontal from ground level. What is its time of flight? ($g = 10\ m/s^2$)

A. 1 s
B. 1.73 s
C. 3.46 s
D. 2 s
E. 4 s

49. A 4 kg box is pulled across a rough floor at constant velocity by a horizontal force of 12 N. What is the coefficient of kinetic friction? ($g = 10\ m/s^2$)

A. 0.3
B. 3
C. 0.4
D. 0.03
E. 0.12

50. A 9 V battery with internal resistance 1 Ω is connected to an external 2 Ω resistor. What is the terminal voltage?

A. 6 V
B. 3 V
C. 4.5 V
D. 9 V
E. 7.5 V

51. The intensity of a sound increases from $10^{-8}\ W/m^2$ to $10^{-6}\ W/m^2$. By how many decibels does the sound level increase?

A. 100 dB
B. 2 dB
C. 10 dB
D. 200 dB
E. 20 dB

52. According to Poiseuille's law, if the radius of a blood vessel is halved (other factors constant), the flow rate becomes:

A. 1/2 of the original.
B. 1/4 of the original.
C. 1/16 of the original.
D. 1/8 of the original.
E. 2 times the original.

53. A satellite orbits Earth in a circular orbit. If its orbital radius were doubled, the gravitational force on it would become:

A. one quarter as large.
B. unchanged.
C. 4 times larger.
D. 2 times larger.
E. half as large.

54. A 0.2 kg sample of metal at 100 °C is placed in 0.4 kg of water at 20 °C. The final temperature is 30 °C. Ignoring losses, what is the specific heat capacity of the metal? ($c_{water} = 4200\ J/(kg\cdot°C)$)

A. 1200 J/(kg·°C)
B. 600 J/(kg·°C)
C. 12 000 J/(kg·°C)
D. 2400 J/(kg·°C)
E. 4200 J/(kg·°C)

55. Solve for x: $\log_3(x + 1) - \log_3(x - 1) = 1$.

A. 4
B. 3
C. 1
D. 2
E. No solution

56. How many solutions does the equation $2 \sin x = 1$ have in the interval $[0, 2\pi)$?

A. 1
B. 0
C. 2
D. 3
E. 4

57. A disease affects 1% of a population. A test detects it in 90% of affected people and gives a false positive in 9% of unaffected people. If a person tests positive, what is the approximate probability that they have the disease?

A. 1%
B. 50%
C. 10%
D. 90%
E. 9%

58. The area enclosed between $y = x^2$ and $y = 4$ is:

A. 64/3
B. 32/3
C. 8
D. 16
E. 16/3

59. For what value(s) of k does the equation $x^2 + kx + 9 = 0$ have exactly one real solution?

A. $k = \pm 6$
B. $k = 3$
C. $k = \pm 3$
D. $k = 9$
E. $k = 6$

60. A radioactive isotope has a decay constant giving a half-life of 8 days. A sample initially has an activity of 3200 Bq. After how many days is the activity 200 Bq?

A. 64 days
B. 32 days
C. 24 days
D. 40 days
E. 16 days

정답 및 해설

번호	정답	해설
1	B	절대위험감소 = 1%p → NNT = 1/0.01 = 100.
2	D	상대위험만 제시하면 실질적 이득이 과장되어 보임.
3	C	멘델.
4	E	고대 그리스.
5	E	졸음 유발 항히스타민제는 금지 → 일부 항히스타민제는 금지.
6	D	교차법: (30−15) : (15−10) = 15 : 5 = 3 : 1.
7	C	시침 105°(90+15), 분침 180° → 75°.
8	E	합 50 → 60, 평균 12. 최댓값 변화는 중앙값 불변 → 8.
9	D	$0.98^2 = 0.9604$.
10	D	만노스-6-인산 표지.
11	D	생식세포·대부분의 암세포.
12	A	방추체 형성 저해 → 중기에서 정지(방추체 점검점).
13	D	전자전달 정지 → 산화적 ATP 생성 중단.
14	B	$X^aY \times X^AX^a$ → 딸: X^aX^A 또는 X^aX^a → 1/2.
15	A	PEP 카복실레이스 → 4탄소 옥살로아세트산.
16	E	Hb 친화력 O_2의 약 200배 이상.
17	B	ATP.
18	E	재분극 지연 → 활동전위 연장.
19	A	동맥 끝: 정수압 > 교질삼투압 → 여과.
20	E	재조합 20% → Ab, aB 각 10%.
21	D	스테로이드는 막 통과, 세포 내 수용체.
22	B	G–C 비율 높을수록 Tm 높음.
23	E	감작된 Rh^- 산모 + Rh^+ 태아.
24	D	헨레 고리 두꺼운 상행지(푸로세마이드).
25	C	감수 2분열 중기: 염색체 2개 × 염색분체 2 = 4.
26	B	비경쟁적 저해: Vmax↓, Km 불변.
27	A	프로모터 → 전사 속도/여부.
28	A	SGLT1: Na^+ 기울기를 이용한 2차 능동수송.
29	E	마이코플라스마는 세포벽 없음.
30	E	CCK.
31	C	한 유전자 → 여러 단백질.
32	D	$q^2 = 0.09$ → q = 0.3, p = 0.7 → 2pq = 0.42 → 420명.
33	B	C:H = 7.14 : 14.3 = 1 : 2 → $CH_2(14) \times 4 = 56$ → C_4H_8.
34	D	$Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$.
35	E	아직 NaOH 0.1 mL분 과량 → 약한 염기성.

번호	정답	해설
36	B	K^+ (18 e^-).
37	B	삼중결합 → sp.
38	B	0.1 mol Mg → 0.1 mol H_2 = 2.24 L.
39	C	3차 알코올은 산화 안 됨.
40	C	pOH 1 → $[OH^-]$ 0.1 × 0.2 L = 0.02 mol × 40 = 0.8 g.
41	B	MgO: 2+/2− 전하, 작은 이온 반경.
42	C	pH = 4.76 + log(1.0/0.1) = 5.76.
43	C	역반응 K = 1/4.
44	A	$2^n = 4$.
45	C	X^{3+} → 13족(예: Al_2O_3).
46	A	묽은 쪽(0.01)이 산화 전극(anode) → 전자는 묽은 쪽 → 진한 쪽.
47	C	글리세롤 + 지방산 나트륨염(비누).
48	D	수직 성분 10 m/s → t = 2×10/10 = 2 s.
49	A	12 = μ×40 → μ = 0.3.
50	A	I = 9/3 = 3 A → 단자전압 3×2 = 6 V.
51	E	10 log(10^2) = 20 dB.
52	C	$Q \propto r^4$ → $(1/2)^4$ = 1/16.
53	A	$F \propto 1/r^2$.
54	A	물이 얻은 열 0.4×4200×10 = 16 800 J = 0.2 × c × 70 → c = 1200.
55	D	(x+1)/(x−1) = 3 → x = 2.
56	C	sin x = 1/2 → π/6, 5π/6.
57	E	양성: 0.009 + 0.0891 = 0.0981 → 0.009/0.0981 ≈ 9%.
58	B	$\int_{-2}^{2} (4 - x^2)\,dx = 16 - 16/3 = 32/3$.
59	A	판별식 $k^2 - 36 = 0$ → k = ±6.
60	B	3200 → 200 = 1/16 → 4 반감기 × 8 = 32일.